
□ MCQs on Data Types (Digital Fundamentals)

1. Which of the following is a floating-point data type?

- A. int
- B. char
- C. float
- D. bool

Ans: C

2. The range of unsigned 8-bit integer is:

- A. 0 to 127
- B. -128 to 127
- C. 0 to 255
- D. -255 to 255

Ans: C

3. A 16-bit signed integer can represent values from:

- A. -32768 to 32767
- B. 0 to 65535
- C. -65536 to 65535
- D. -128 to 127

Ans: A

4. What is the size of a float in most modern 32-bit systems?

- A. 2 bytes
- B. 4 bytes
- C. 6 bytes
- D. 8 bytes

Ans: B

5. Which of the following is not a basic data type in C?

- A. int
- B. struct
- C. char
- D. float

Ans: B

6. A Boolean data type can have how many values?

- A. 0
- B. 1

C. 2

D. Infinite

Ans: C

7. In C, the data type used for single character storage is:

A. char

B. string

C. int

D. byte

Ans: A

8. What is the precision of a double data type?

A. 4 decimal digits

B. 6 decimal digits

C. 10 decimal digits

D. 15-16 decimal digits

Ans: D

9. Which data type is best for storing a person's age?

A. float

B. char

C. int

D. bool

Ans: C

10. Which of the following is not a valid C data type?

A. double

B. float

C. real

D. int

Ans: C

□ CUET PG / MCA PYQ Style Questions

11. (CUET PG) The range of an 8-bit signed integer is:

A. -128 to 127

B. 0 to 255

C. -255 to 255

D. -127 to 128

Ans: A

12. (PYQ) The size of a double variable is generally:

- A. 2 bytes
- B. 4 bytes
- C. 6 bytes
- D. 8 bytes

Ans: D

13. (CUET PG-style) Which of the following is a derived data type?

- A. int
- B. float
- C. struct
- D. char

Ans: C

14. Which of the following can hold decimal values?

- A. int
- B. char
- C. float
- D. bool

Ans: C

15. Which data type is suitable for storing status flags (true/false)?

- A. bool
- B. float
- C. int
- D. char

Ans: A

More Practice

16. How many values can a 3-bit binary number represent?

- A. 8
- B. 6
- C. 16
- D. 4

Ans: A

17. A short int in C typically uses:

- A. 8 bits
- B. 16 bits

- C. 32 bits
- D. 64 bits

Ans: B

18. What is the smallest standard data type in C?

- A. int
- B. char
- C. double
- D. float

Ans: B

19. The C data type unsigned int means:

- A. No sign is used
- B. It stores only positive values
- C. Twice the normal range
- D. Stores only characters

Ans: B

20. Data type used to store large numbers accurately in scientific calculations:

- A. int
- B. float
- C. double
- D. char

Ans: C

Final 10 Questions

21. Which of these types occupies the most memory?

- A. int
- B. float
- C. double
- D. char

Ans: C

22. Data type long in C has a size of:

- A. 2 bytes
- B. 4 bytes
- C. 8 bytes
- D. System dependent

Ans: D

23. A 64-bit processor means it can handle:

- A. 64 instructions at once
- B. 64-bit wide memory addresses
- C. 64-bit graphics
- D. None

Ans: B

24. Which of the following is not a scalar data type?

- A. int
- B. float
- C. array
- D. char

Ans: C

25. In Java, byte data type has range:

- A. 0 to 255
- B. -128 to 127
- C. -255 to 255
- D. 0 to 127

Ans: B

26. Which data type supports decimal and exponential representation?

- A. int
- B. float
- C. char
- D. bool

Ans: B

27. The default data type for decimal values in C is:

- A. double
- B. int
- C. float
- D. long

Ans: A

28. Which data type is best suited to store 'Y' or 'N' values?

- A. int
- B. char
- C. bool
- D. float

Ans: B

29. The data type used to store 1 character in C is:

- A. int
- B. char
- C. string
- D. bool

Ans: B

30. (CUET PYQ Style) What is the size of bool in C++?

- A. 1 byte
- B. 2 bytes
- C. 4 bytes
- D. 8 bytes

Ans: A

MCQ Set on Number Systems and Conversion:

☐ CUET PG MCA – Number Systems & Conversions: 30 MCQs

Q1. What is the binary equivalent of the decimal number 25?

- A. 11001
- B. 11101
- C. 10101
- D. 10011

☐ **Answer:** A

Q2. What is the decimal equivalent of the binary number 11010?

- A. 24
- B. 25
- C. 26
- D. 30

☐ **Answer:** C

Q3. Which of the following number systems is base-8?

- A. Binary
- B. Decimal
- C. Octal
- D. Hexadecimal

☐ **Answer:** C

Q4. Convert 11110011 (binary) to hexadecimal.

- A. F3
- B. E3
- C. D2
- D. C4

☐ **Answer:** A

Q5. What is the binary equivalent of the hexadecimal number A7?

- A. 10100111
- B. 11101001
- C. 11010110
- D. 10111001

☐ **Answer:** A

Q6. What is the hexadecimal representation of the decimal number 255?

- A. FF
- B. FE
- C. F0
- D. EF

☐ **Answer:** A

Q7. Convert the octal number 75 to decimal.

- A. 61
- B. 62
- C. 65
- D. 69

☐ **Answer:** C

Q8. Which of the following is a valid binary number?

- A. 1021
- B. 0101
- C. 238
- D. 8B3

☐ **Answer:** B

Q9. Convert the decimal number 64 into octal.

- A. 80
- B. 100
- C. 120
- D. 144

☐ **Answer:** B

Q10. Which number system uses digits 0 to 9 and A to F?

- A. Decimal
- B. Octal
- C. Binary
- D. Hexadecimal

☐ **Answer:** D

Q11. The number 0x1F in hexadecimal is equivalent to ____ in decimal.

- A. 15
- B. 25
- C. 31
- D. 41

☐ **Answer:** C

Q12. Which of the following is NOT a positional number system?

- A. Decimal
- B. Roman
- C. Binary
- D. Octal

☐ **Answer:** B

Q13. Which base is used in the decimal number system?

- A. 2
- B. 8
- C. 10
- D. 16

☐ **Answer:** C

Q14. Convert decimal 345 to hexadecimal.

- A. 159
- B. 153
- C. 158
- D. 15F

☐ **Answer:** D

Q15. What is 1001 in decimal?

- A. 7
- B. 9
- C. 11
- D. 10

☐ **Answer:** B

Q16. Convert hexadecimal 2A to binary.

- A. 101110
- B. 101001
- C. 00101010
- D. 100011

☐ **Answer:** C

Q17. Octal 123 is equivalent to binary:

- A. 1010011
- B. 01001011
- C. 001010011
- D. 110110

☐ **Answer:** C

Q18. Decimal 1000 = ____ in binary

- A. 1111101000
- B. 1001001000
- C. 111101000
- D. 111110000

☐ **Answer:** A

Q19. Binary to decimal: 101101

- A. 45
- B. 43
- C. 47
- D. 41

☐ **Answer:** A

Q20. What is the hexadecimal for binary 1101 0101?

- A. D4
- B. D5
- C. A7
- D. E3

☐ **Answer:** B

☐ **CUET PG PYQs from this topic:**

Q21. CUET PG 2023: Which of the following represents the octal equivalent of binary 101011?

- A. 53
- B. 56

C. 52

D. 54

☐ **Answer: A**

Q22. CUET PG 2022: Convert hexadecimal 3F into decimal.

A. 62

B. 63

C. 64

D. 65

☐ **Answer: B**

Q23. CUET PG 2021: What is the binary equivalent of decimal number 255?

A. 11111110

B. 11111100

C. 11111111

D. 11111000

☐ **Answer: C**

More Practice (24-30):

Q24. What is the decimal equivalent of octal 17?

A. 15

B. 16

C. 17

D. 18

☐ **Answer: B**

Q25. Hexadecimal 7F equals what in binary?

A. 01111110

B. 01111111

C. 00111111

D. 11111111

☐ **Answer: B**

Q26. Which of the following number systems is NOT used in computers?

A. Binary

B. Hexadecimal

C. Quaternary

D. Decimal

☐ **Answer: C**

Q27. Which number system is used by digital computers?

- A. Octal
- B. Hexadecimal
- C. Binary
- D. Decimal

☐ **Answer:** C

Q28. Convert binary 100000 into decimal.

- A. 64
- B. 32
- C. 16
- D. 128

☐ **Answer:** B

Q29. Convert decimal 18 to binary.

- A. 10010
- B. 10101
- C. 10001
- D. 10011

☐ **Answer:** A

Q30. Convert hexadecimal 1A to decimal.

- A. 20
- B. 25
- C. 26
- D. 30

☐ **Answer:** C

☐ **Topic:** Complements (1's and 2's Complement, 9's and 10's Complement, r's and (r-1)'s Complement)

1. The 1's complement of binary number 101100 is:

- a) 010011
 - b) 101100
 - ☐ **c) 010011**
 - d) 101011
-

2. The 2's complement of binary number 1101 is:

- a) 1100
 - ☒ b) 0011
 - c) 1110
 - d) 0001
-

3. The 2's complement of 00001101 is:

- a) 00001100
 - b) 00001111
 - ☒ c) 11110011
 - d) 00010010
-

4. The 2's complement of 0 is:

- a) 0
 - ☒ b) 0
 - c) 1
 - d) Undefined
-

5. What is the 1's complement of 11111111?

- ☒ a) 00000000
 - b) 11111110
 - c) 00000001
 - d) 11111111
-

6. The 9's complement of decimal number 456 is:

- a) 654
 - ☒ b) 543
 - c) 999
 - d) 444
-

7. The 10's complement of 456 is:

- a) 543
 - ☒ b) 544
 - c) 555
 - d) 545
-

8. The 9's complement of 1000 is:

- a) 1000
 - ☒ b) 8999
 - c) 9999
 - d) 0001
-

9. 2's complement of binary number 00000101 is:

- a) 00000101
 - ☒ b) 11111011
 - c) 00001011
 - d) 11111111
-

10. r's complement is used mainly for:

- ☒ a) Subtraction
 - b) Multiplication
 - c) Addition
 - d) Division
-

11. The $(r-1)$'s complement of decimal number 327 is:

- a) 672
 - ☒ b) 672
 - c) 773
 - d) 888
-

12. In base-10 system, $(r-1)$'s complement is known as:

- a) 1's complement
 - ☒ b) 9's complement
 - c) 10's complement
 - d) 8's complement
-

13. Which method uses 2's complement for subtraction in binary arithmetic?

- ☒ a) Complement and add method
 - b) Direct subtraction
 - c) Decimal conversion
 - d) Floating subtraction
-

14. 2's complement representation is used in:

- a) BCD arithmetic
 - ☒ b) Signed binary arithmetic
 - c) Hexadecimal multiplication
 - d) Boolean algebra
-

15. 1's complement of 00000000 is:

- ☒ a) 11111111
 - b) 00000001
 - c) 00000000
 - d) 11111110
-

16. The 10's complement of 999 is:

- a) 999
 - b) 000
 - ☒ c) 001
 - d) 1001
-

17. Which of the following is true about 2's complement?

- ☐ a) It represents negative numbers
 - b) It's used only in floating point
 - c) Cannot represent zero
 - d) Same as 1's complement
-

18. What is the 2's complement of 00000001?

- a) 11111110
 - ☐ b) 11111111
 - c) 00000000
 - d) 11111101
-

19. The 9's complement of 1234 is:

- a) 8765
 - ☐ b) 8765
 - c) 8766
 - d) 8764
-

20. Which of the following is NOT true for 2's complement?

- a) Overflow can be detected
 - b) No ambiguity of ± 0
 - ☐ c) Double representation of zero
 - d) Used for subtraction
-

21. 2's complement of 11111101 is:

- a) 00000001
 - ☐ b) 00000011
 - c) 00000010
 - d) 11111111
-

22. In 2's complement form, -128 (8 bits) is represented as:

- ☐ a) 10000000
 - b) 11111111
 - c) 01111111
 - d) 00000000
-

23. 1's complement of 00110011 is:

- a) 11001100
 - ☐ b) 11001100
 - c) 11111111
 - d) 00001111
-

24. How is 0 represented in 2's complement (8-bit)?

- ☐ a) 00000000
 - b) 11111111
 - c) 10000000
 - d) 11110000
-

25. 2's complement of 01010101 is:

- a) 10101010
 - ☐ b) 10101011
 - c) 01010110
 - d) 10101000
-

26. What is the decimal equivalent of 2's complement number 11111100?

- a) 4
 - ☐ b) -4
 - c) -2
 - d) -3
-

27. The result of 9's complement of 0751 is:

- a) 9248
 - ☒ b) 9248
 - c) 8749
 - d) 9250
-

28. Which complement method avoids end-around carry?

- ☒ a) 2's complement
 - b) 1's complement
 - c) 9's complement
 - d) r-1 complement
-

29. What is the 2's complement of 11111111?

- a) 11111111
 - ☒ b) 00000001
 - c) 11111110
 - d) 00000000
-

30. (PYQ) CUET PG: The 2's complement of binary number 0101 is:

- a) 1010
 - ☒ b) 1011
 - c) 1111
 - d) 0001
-

☒ **Topic: Fixed Point Representation**

1. In fixed point representation, the position of the binary point is:

- ☒ a) Fixed
 - b) Variable
 - c) Floating
 - d) User defined
-

2. Fixed point representation is mainly used to represent:

- a) Very large numbers
 - ☒ b) Integers and limited precision decimals
 - c) Complex numbers
 - d) Imaginary numbers
-

3. In signed fixed-point binary, negative numbers are usually represented using:

- a) 1's complement
 - b) Sign-magnitude
 - ☒ c) 2's complement
 - d) BCD
-

4. The integer part and fractional part are separated by:

- a) Dot sign
 - b) Hyphen
 - ☒ c) Binary point
 - d) Exponent
-

5. A 16-bit fixed-point number with 8 bits for fraction can represent:

- a) Only integers
 - ☒ b) 8 integer bits and 8 fractional bits
 - c) 16 fractional bits
 - d) Negative exponents only
-

6. Fixed-point format is best suited for:

- a) Scientific computing
 - ☒ b) Embedded systems with limited resources
 - c) Deep learning
 - d) Operating systems
-

7. A number in Q8.8 format means:

- a) 8 fractional bits only
 - ☒ b) 8 integer bits and 8 fractional bits
 - c) 16 fractional bits
 - d) 8 floating bits
-

8. Which of the following has higher precision?

- a) 8.8 format
 - ☒ b) 4.12 format
 - c) 10.6 format
 - d) 12.4 format
-

9. In fixed-point representation, overflow happens when:

- a) Fraction is too long
 - b) Sign bit is lost
 - ☒ c) Value exceeds the representable range
 - d) Binary point is shifted
-

10. The fixed point representation of 5.25 in 4.4 format is:

- ☒ a) 0101.0100
 - b) 0101.0010
 - c) 0100.1010
 - d) 0001.1100
-

11. What is the maximum unsigned value for 8-bit fixed-point with 4 fractional bits?

- ☒ a) 15.9375
 - b) 31.875
 - c) 7.5
 - d) 8.5
-

12. In fixed-point arithmetic, truncation error occurs due to:

- ☒ a) Limited number of fractional bits
 - b) Overflow
 - c) Division
 - d) Integer rounding
-

13. Which is NOT an advantage of fixed-point representation?

- a) Faster arithmetic
 - b) Low power use
 - ☒ c) Dynamic range
 - d) Simple hardware
-

14. Fixed-point numbers are generally stored as:

- ☒ a) Integers with implied binary point
 - b) Floating-point words
 - c) Strings
 - d) Exponential forms
-

15. If the binary number is 00001010, and format is 4.4, what is the value?

- a) 10
 - ☒ b) 0.625
 - c) 5.5
 - d) 2.5
-

16. In Q3.5 format, the range of numbers is:

- a) -8 to +7.5
 - ☒ b) -8.0 to +7.96875
 - c) -4.0 to +3.96875
 - d) -7.5 to +7.5
-

17. Which operation is more complex in fixed-point than floating-point?

- a) Addition
 - ☒ b) **Scaling and normalization**
 - c) Bit shift
 - d) Sign detection
-

18. In 8-bit signed fixed-point (Q4.4), the smallest positive number is:

- a) 0.25
 - ☒ b) **0.0625**
 - c) 0.125
 - d) 1.0
-

19. The Q number format is used to define:

- a) Floating point
 - ☒ b) **Fixed-point binary representation**
 - c) ASCII values
 - d) Address format
-

20. Which of the following is not a valid Q format?

- a) Q15.0
 - b) Q0.15
 - ☒ c) **Q8.16 in 16-bit register**
 - d) Q7.8
-

21. What is the decimal equivalent of Q2.6 number 11001000?

- a) -2.25
 - ☒ b) **-1.5**
 - c) -3.75
 - d) -1.75
-

22. In fixed-point, the binary point:

- a) Is stored separately
 - ☒ b) Is implied by format
 - c) Varies at runtime
 - d) Depends on exponent
-

23. For fixed-point Q5.3 format, which decimal value does 10111001 represent?

- a) -6.125
 - ☒ b) -5.875
 - c) -4.625
 - d) -7.125
-

24. Which of the following causes round-off errors in fixed-point?

- a) Overflow
 - ☒ b) Limited fraction bits
 - c) Format change
 - d) Addition
-

25. Fixed-point multiplication needs:

- a) Decimal alignment
 - ☒ b) Binary point adjustment after multiply
 - c) Bitwise shift left
 - d) Division
-

26. (PYQ) CUET PG: In fixed-point Q3.5 format, which of the following values can be exactly represented?

- a) 0.2
 - ☒ b) 0.5
 - c) 0.3
 - d) 0.7
-

27. For 8-bit Q4.4 fixed-point, what is the binary for -2.5?

- a) 11101000
 - ☒ b) 11101000
 - c) 00010100
 - d) 00101000
-

28. How many unique values can be represented in Q4.4 signed fixed-point format?

- a) 16
 - b) 64
 - ☒ c) 256
 - d) 128
-

29. What is the value of binary 01011100 in Q4.4 format?

- a) 5.5
 - b) 4.25
 - ☒ c) 5.75
 - d) 3.5
-

30. Why is fixed-point preferred in real-time embedded systems?

- a) Floating-point is easier
 - b) High-level math needed
 - ☒ c) Speed and lower hardware complexity
 - d) No rounding errors
-

Here are **30 MCQs on Floating Point Representation** (under Digital Fundamentals), including **CUET PG Previous Year Questions (PYQs)** wherever available.

☒ **Topic: Floating Point Representation**

☒ Total: 30 MCQs

☒ Correct answers marked

☒ CUET PG PYQs tagged as **(PYQ)**

1. Floating-point representation is used to represent:

- a) Only integers
 - ☒ **b) Real numbers with large range**
 - c) Imaginary numbers
 - d) ASCII codes
-

2. A normalized floating-point number always has:

- a) All 1s in exponent
 - ☒ **b) 1 as the most significant bit of the mantissa**
 - c) 0 in exponent
 - d) 0.5 in mantissa
-

3. What is the bias value for 8-bit exponent in IEEE 754 single-precision format?

- a) 127
 - ☒ **b) 127**
 - c) 255
 - d) 128
-

4. In IEEE 754 single-precision format, the number of bits used for mantissa is:

- ☒ **a) 23**
 - b) 8
 - c) 31
 - d) 24
-

5. In IEEE 754 double-precision, the exponent field is:

- a) 8 bits
 - ☒ **b) 11 bits**
 - c) 16 bits
 - d) 32 bits
-

6. What does the term denormalized number mean in floating-point?

- ☐ a) Mantissa does not begin with 1
 - b) Zero exponent
 - c) Overflowed value
 - d) NaN
-

7. Which of the following values indicates positive infinity in IEEE 754 format?

- a) Sign = 1, Exponent = 255, Mantissa = 0
 - ☐ b) Sign = 0, Exponent = 255, Mantissa = 0
 - c) Sign = 0, Exponent = 0, Mantissa = 0
 - d) Exponent = 127
-

8. Floating-point representation suffers mainly from:

- a) Overflow
 - b) Underflow
 - ☐ c) Precision loss and rounding error
 - d) Noise
-

9. In IEEE 754 single precision, how many bits are there in total?

- a) 32
 - ☐ b) 32
 - c) 16
 - d) 64
-

10. (PYQ) CUET PG: The decimal number -0.75 is represented in binary as:

- a) -1.01
 - ☐ b) -0.11
 - c) -0.10
 - d) -1.10
-

11. What is the sign bit for negative numbers in IEEE floating point?

- a) 0
 - ☒ b) 1
 - c) Depends on mantissa
 - d) Depends on exponent
-

12. Which part of the floating-point number determines the magnitude?

- a) Sign
 - b) Mantissa
 - ☒ c) Exponent
 - d) Parity
-

13. The smallest positive normalized number in single precision is:

- ☒ a) 2^{-126}
 - b) 2^{-127}
 - c) 2^{-255}
 - d) 0
-

14. Floating-point numbers are generally expressed in:

- a) Sign-Magnitude format
 - b) BCD
 - ☒ c) Scientific notation
 - d) Fixed decimal
-

15. (PYQ) CUET PG: What does NaN represent in IEEE floating point format?

- a) Negative answer
 - ☒ b) Not a Number
 - c) Normal and Negative
 - d) No actual number
-

16. Which of the following is true for IEEE 754 special case zero?

- a) Mantissa = 1
 - b) Exponent = 127
 - ☒ c) Exponent = 0, Mantissa = 0
 - d) Sign = 1 only
-

17. Double precision allows representation of:

- a) Smaller values only
 - ☒ b) Higher precision and wider range
 - c) Integers only
 - d) ASCII values
-

18. In IEEE format, what does Exponent = 255 and Mantissa $\neq$ 0 indicate?

- a) Positive infinity
 - b) Zero
 - ☒ c) NaN
 - d) Denormalized number
-

19. Which is NOT a valid IEEE 754 representation?

- a) +0
 - ☒ b) +infinity with non-zero mantissa
 - c) -0
 - d) +NaN
-

20. Which of the following is most accurate?

- a) Fixed-point Q format
 - ☒ b) IEEE 754 double precision
 - c) Sign-magnitude format
 - d) 1's complement
-

21. What does mantissa determine?

- a) Sign
 - b) Exponent
 - ☒ c) Precision
 - d) Range
-

22. Range of exponent in IEEE single-precision is:

- a) -128 to +127
 - ☒ b) -126 to +127
 - c) -255 to +255
 - d) 0 to 255
-

23. The decimal number 6.25 is represented in binary as:

- a) 101.11
 - ☒ b) 110.01
 - c) 110.11
 - d) 111.01
-

24. In normalized floating-point, the hidden bit is:

- ☒ a) Always 1
 - b) Always 0
 - c) Based on exponent
 - d) Based on sign
-

25. Rounding in floating-point arithmetic may lead to:

- a) Memory loss
 - ☒ b) Precision loss
 - c) Logical error
 - d) Bit corruption
-

26. What is the bias used in IEEE 754 double precision?

- a) 1023
 - ☒ b) 1023
 - c) 127
 - d) 255
-

27. In IEEE 754, what is the base (radix)?

- a) 2
 - ☒ b) 2
 - c) 10
 - d) 16
-

28. The largest positive number in IEEE 754 single-precision is approximately:

- ☒ a) 3.4×10^{38}
 - b) 1.7×10^{308}
 - c) 2^{75}
 - d) 255.0
-

29. Denormalized numbers are used to represent:

- ☒ a) Very small values close to zero
 - b) Negative zero
 - c) Large exponents
 - d) NaN
-

30. (PYQ) CUET PG: The purpose of biasing in floating point is to:

- a) Reduce exponent
 - ☒ b) Represent both positive and negative exponents
 - c) Shift mantissa
 - d) Normalize sign
-

Here are **30 MCQs** on **Error Detection Codes** (under Digital Fundamentals), including **CUET PG Previous year questions (PYQs)** wherever available.

☐ **Topic: Error Detection Codes**

☐ Total: 30 MCQs

☐ Correct answers are marked

☐ CUET PG PYQs are tagged as **(PYQ)**

1. Which of the following is used for error detection?

- a) Gray Code
 - b) Hamming Code
 - ☐ **c) Parity Bit**
 - d) Excess-3 Code
-

2. A parity bit is:

- ☐ **a) An extra bit added for error detection**
 - b) A type of logical gate
 - c) Used to speed up computation
 - d) Part of the data
-

3. Parity checking can detect:

- ☐ **a) Single-bit errors**
 - b) All multiple-bit errors
 - c) No errors
 - d) Only even bit errors
-

4. Even parity for binary 1011 is:

- a) 0
 - ☐ **b) 1**
 - c) 2
 - d) 3
-

5. Odd parity for 10000000 is:

- ☒ a) 1
 - b) 0
 - c) 8
 - d) Cannot be determined
-

6. What is the drawback of parity checking?

- a) Too slow
 - ☒ b) Cannot detect even number of bit errors
 - c) Expensive
 - d) Requires large memory
-

7. Hamming code can detect and correct:

- a) Only errors
 - b) Only detection
 - ☒ c) Single-bit errors
 - d) Multiple-bit errors
-

8. The minimum number of redundant bits (r) needed to detect and correct 1-bit errors in 7-bit data is:

- a) 2
 - ☒ b) 4
 - c) 5
 - d) 6
-

9. CRC stands for:

- ☒ a) Cyclic Redundancy Check
 - b) Circular Record Code
 - c) Coding Repetition Check
 - d) Current Repeat Circuit
-

10. Which error detection method is used in network protocols like Ethernet?

- a) Hamming
 - b) Parity
 - ☒ c) CRC
 - d) Checksum
-

11. Which is the most powerful error detection technique?

- a) Parity
 - b) Checksum
 - ☒ c) CRC
 - d) Repetition
-

12. The polynomial used in CRC is called:

- ☒ a) Generator Polynomial
 - b) Correction Polynomial
 - c) Verification Key
 - d) Base Equation
-

13. Which of the following cannot correct errors?

- ☒ a) Parity
 - b) Hamming
 - c) Repetition
 - d) ECC
-

14. Checksum is widely used in:

- a) Video encoding
 - ☒ b) Data transmission protocols
 - c) Floating-point unit
 - d) CPU arithmetic
-

15. In even parity, if the number of 1's is odd, the parity bit should be:

- ☒ a) 1
 - b) 0
 - c) Don't care
 - d) Equal to number of 0's
-

16. Two-dimensional parity can detect:

- a) Only single-bit errors
 - ☒ b) Single and some double-bit errors
 - c) Only even errors
 - d) Only odd errors
-

17. Error detection is important for:

- ☒ a) Reliable data communication
 - b) Audio output
 - c) OS scheduling
 - d) Power reduction
-

18. Which is NOT an error detection method?

- a) Parity
 - b) CRC
 - ☒ c) Binary Weighted Code
 - d) Checksum
-

19. (PYQ) CUET PG: Parity checking can detect:

- ☒ a) All single-bit errors
 - b) All errors
 - c) No errors
 - d) Only double-bit errors
-

20. Which code uses modulo-2 division?

- a) Hamming
 - ☒ b) CRC
 - c) Checksum
 - d) Odd parity
-

21. One-bit error correction can be achieved using:

- a) Even parity
 - ☒ b) Hamming code
 - c) CRC
 - d) Checksum
-

22. A triple repetition code increases data by:

- a) 50%
 - b) 100%
 - ☒ c) 200%
 - d) 300%
-

23. In CRC, which operation is used?

- a) AND
 - b) OR
 - ☒ c) XOR
 - d) NOT
-

24. (PYQ) CUET PG: If data = 1101 and generator = 1011, what's the degree of polynomial used?

- a) 4
 - ☒ b) 3
 - c) 2
 - d) 5
-

25. Which bit combination causes even parity failure?

- a) 1100 with parity 0
 - ☒ b) 1100 with parity 1
 - c) 1001 with parity 0
 - d) 1000 with parity 0
-

26. Hamming code works only when:

- ☒ a) Single bit error occurs
 - b) Two bits are wrong
 - c) 3-bit errors occur
 - d) Data is floating point
-

27. Which of the following is not used for error detection?

- a) Checksum
 - b) CRC
 - ☒ c) Floating point
 - d) Parity
-

28. What is the goal of error detection codes?

- a) Generate unique values
 - b) Encrypt data
 - ☒ c) Identify transmission errors
 - d) Format the disk
-

29. Error detection codes are inserted:

- a) In RAM only
 - b) After transmission
 - ☒ c) Before and during transmission
 - d) In cache
-

30. A 7-bit Hamming code can encode:

- a) 4 data bits
 - ☒ b) 4 data bits + 3 parity bits
 - c) 7 data bits
 - d) 3 data bits + 4 parity bits
-

Here are **30 MCQs** on **Computer Arithmetic - Addition** (under Digital Fundamentals), including **CUET PG Previous Year Questions (PYQs)** wherever applicable.

☐ **Topic: Computer Arithmetic - Addition**

- ☐ Total: 30 MCQs
 - ☐ Correct answers marked
 - ☐ CUET PG PYQs are marked as (PYQ)
-

1. In binary addition, $1 + 1 =$

- a) 1
 - b) 0
 - ☒ c) 10
 - d) 11
-

2. In binary, $1 + 1 + 1$ equals:

- a) 10
 - ☒ b) 11
 - c) 111
 - d) 01
-

3. Binary addition is similar to:

- a) Octal addition
 - ☒ b) Decimal addition using base 2
 - c) ASCII operation
 - d) Gray code conversion
-

4. Half adder adds:

- ☒ a) Two single-bit numbers
 - b) Two multi-bit numbers
 - c) One signed number
 - d) One BCD number
-

5. The output of a half adder is:

- a) Only sum
 - b) Only carry
 - ☒ c) Sum and Carry
 - d) Only difference
-

6. A full adder adds:

- ☒ a) Three bits
 - b) Two bits
 - c) One bit
 - d) Four bits
-

7. The carry output of full adder is '1' when:

- ☒ a) At least two inputs are 1
 - b) All inputs are 0
 - c) Only one input is 1
 - d) All inputs are different
-

8. What is the sum of binary numbers 1101 and 1011?

- ☒ a) 11000
 - b) 10100
 - c) 11110
 - d) 10001
-

9. What logic gate represents binary addition (SUM) in half adder?

- a) AND
 - ☒ b) XOR
 - c) NAND
 - d) OR
-

10. Carry in binary addition is generated when:

- ☒ a) Both bits are 1
 - b) Any bit is 0
 - c) All bits are 0
 - d) All bits are different
-

11. How many full adders are required to add two 4-bit numbers?

- a) 1
 - b) 2
 - ☒ c) 4
 - d) 3
-

12. In ripple carry adder, the carry is:

- ☒ a) Propagated through each stage
 - b) Reset after each sum
 - c) Always zero
 - d) Not used
-

13. In signed binary addition, overflow occurs when:

- a) One input is negative
 - ☒ b) Sum exceeds representable range
 - c) Carry is zero
 - d) XOR gate fails
-

14. Binary addition $1010 + 0101 = ?$

- a) 1010
 - b) 0110
 - ☒ c) 1111
 - d) 1101
-

15. The sum of 1111 and 0001 in binary is:

- ☒ a) 10000
 - b) 11110
 - c) 0000
 - d) 1111
-

16. What is the sum of 1001 + 0110?

- a) 1111
 - ☒ b) 1111
 - c) 1101
 - d) 0011
-

17. (PYQ) CUET PG: The result of adding 1011 and 1101 is:

- a) 10100
 - ☒ b) 11000
 - c) 11111
 - d) 10001
-

18. What is the carry and sum in half adder for inputs A=1, B=1?

- a) Carry=0, Sum=1
 - ☒ b) Carry=1, Sum=0
 - c) Carry=1, Sum=1
 - d) Carry=0, Sum=0
-

19. A full adder can be built using:

- a) 2 NAND gates
 - b) 2 half subtractors
 - ☒ c) 2 half adders and an OR gate
 - d) 1 half adder only
-

20. Overflow in addition indicates:

- a) Memory corruption
 - ☒ b) Incorrect sign in result
 - c) Program crash
 - d) Register loss
-

21. In binary addition, sum of 111 and 001 is:

- a) 1000
 - ☒ b) 1000
 - c) 111
 - d) 0110
-

22. What is the decimal value of binary addition: 0110 + 0011?

- a) 7
 - ☒ b) 9
 - c) 6
 - d) 10
-

23. If two 4-bit signed numbers are added and result is 5 bits, what occurred?

- a) Subtraction
 - ☒ b) Overflow
 - c) Underflow
 - d) Bit masking
-

24. Which adder eliminates delay due to carry propagation?

- ☒ a) Carry look-ahead adder
 - b) Ripple carry adder
 - c) Full adder
 - d) BCD adder
-

25. What is the result of $1011 + 0111$?

- ☒ a) 10010
 - b) 10110
 - c) 01110
 - d) 11000
-

26. If $A=1$, $B=0$, $C_{in}=1$ in full adder, what is Sum and Carry?

- a) Sum=0, Carry=0
 - ☒ b) Sum=0, Carry=1
 - c) Sum=1, Carry=0
 - d) Sum=1, Carry=1
-

27. What logic gate is used for carry generation in half adder?

- a) XOR
 - ☒ b) AND
 - c) OR
 - d) NAND
-

28. Full adder outputs are:

- ☒ a) Sum and Carry
 - b) Difference and Borrow
 - c) Product and Quotient
 - d) Only Sum
-

29. Which adder is most efficient for high-speed arithmetic?

- a) Full adder
 - ☒ b) Carry look-ahead adder
 - c) Ripple adder
 - d) Serial adder
-

30. In BCD addition, when sum > 1001, what is added?

- a) 0000
 - ☒ b) 0110 (6)
 - c) 1111
 - d) 1000
-

Here are **30 MCQs** on **Computer Arithmetic – Multiplication** (under Digital Fundamentals), including **CUET PG Previous Year Questions (PYQs)** wherever available.

☒ **Topic: Computer Arithmetic – Multiplication**

- ☒ Total: 30 MCQs
 - ☒ Correct answers are marked
 - ☒ CUET PG PYQs are tagged as **(PYQ)**
-

1. Binary multiplication is similar to:

- a) Binary addition
 - b) Decimal addition
 - ☒ c) Decimal multiplication
 - d) Hexadecimal subtraction
-

2. In binary, $1 \times 1 = ?$

- a) 0
 - b) 2
 - ☒ c) 1
 - d) 10
-

3. In binary multiplication, any number multiplied by 0 gives:

- ☒ a) 0
 - b) 1
 - c) Same number
 - d) Overflow
-

4. Multiplying 101×11 in binary gives:

- a) 1000
 - ☒ b) 1111
 - c) 1001
 - d) 1101
-

5. Binary multiplication has how many basic operations?

- ☒ a) 4
 - b) 2
 - c) 3
 - d) 5
-

6. The partial product in binary multiplication is formed by:

- a) Addition
 - ☒ b) ANDing with each bit
 - c) XOR operation
 - d) Subtraction
-

7. What is the result of binary multiplication: 1001×0011 ?

- a) 1110
 - b) 1010
 - ☒ c) 11011
 - d) 1111
-

8. (PYQ) CUET PG: Binary multiplication of $111 \times 10 = ?$

- a) 111
 - ☒ b) 1110
 - c) 1010
 - d) 1100
-

9. Booth's algorithm is used for:

- a) Addition
 - ☒ b) Signed binary multiplication
 - c) Division
 - d) Error correction
-

10. In Booth's multiplication algorithm, the operation depends on:

- ☒ a) Two LSBs of the multiplier
 - b) MSB of multiplicand
 - c) All bits
 - d) Carry flag
-

11. What is the key advantage of Booth's algorithm?

- a) Hardware simplicity
 - ☒ b) Efficient handling of negative numbers
 - c) Fast subtraction
 - d) CRC generation
-

12. In Booth's algorithm, 10 means:

- a) Do nothing
 - b) Add multiplicand
 - ☒ c) Subtract multiplicand
 - d) Set overflow
-

13. What is 2's complement of 0101?

- a) 1010
 - b) 1111
 - ☒ c) 1011
 - d) 0110
-

14. Unsigned binary multiplication of 6×5 equals:

- ☒ a) 11110
 - b) 10101
 - c) 10000
 - d) 10001
-

15. If $A = 1001$ and $B = 0101$, what is $A \times B$?

- a) 11100
 - b) 11010
 - ☒ c) 11001
 - d) 10110
-

16. Multiplying by 2 in binary is equivalent to:

- ☒ a) Left shift by 1
 - b) Right shift by 1
 - c) Add 1
 - d) Complement
-

17. Multiplying by 4 in binary equals:

- a) Add 4
 - ☒ b) Left shift by 2
 - c) AND with 100
 - d) Subtract 4
-

18. Wallace Tree is used in:

- a) Addition
 - ☒ b) High-speed multiplication
 - c) Division
 - d) Encoding
-

19. A multiplication overflow occurs when:

- a) Carry is not zero
 - ☒ b) Result exceeds fixed word size
 - c) Product is zero
 - d) Bits are equal
-

20. Carry-save adders are used in:

- a) Parity logic
 - b) Subtraction
 - ☒ c) Fast multipliers
 - d) Comparators
-

21. In an array multiplier, number of AND gates required for $n \times n$ bits is:

- a) n
 - b) n^2
 - ☒ c) n^2
 - d) $2n$
-

22. The time complexity of basic binary multiplication algorithm is:

- a) $O(1)$
 - b) $O(n)$
 - ☒ c) $O(n^2)$
 - d) $O(\log n)$
-

23. Which algorithm performs multiplication using shift and add method?

- a) Divide-and-conquer
 - ☒ b) Shift-add algorithm
 - c) CRC
 - d) Newton-Raphson
-

24. In signed multiplication, sign of result is positive if:

- a) One operand is positive
 - ☒ b) Both operands have same sign
 - c) One operand is zero
 - d) Always negative
-

25. A 4-bit multiplier can multiply numbers up to:

- ☒ a) 15×15
 - b) 16×16
 - c) 31×31
 - d) 8×8
-

26. Which multiplier is fastest in general?

- a) Booth multiplier
 - ☒ b) Wallace tree multiplier
 - c) Array multiplier
 - d) Ripple carry multiplier
-

27. For 4-bit inputs, maximum product size is:

- a) 4 bits
 - b) 6 bits
 - ☒ c) 8 bits
 - d) 16 bits
-

28. The output of multiplying 2's complement numbers 1100×0010 :

- a) 11111100
 - ☒ b) 11111000
 - c) 00000100
 - d) 00001100
-

29. In signed multiplication, negative $\times$ positive = ?

- a) Positive
 - ☒ b) Negative
 - c) Zero
 - d) Overflow
-

30. Multiplication in computer is usually performed using:

- a) Subtraction
 - b) Repeated XOR
 - ☒ c) Shift and add
 - d) Memory mapping
-

Here are **30 MCQs** on **Computer Arithmetic – Division Algorithms** (under Digital Fundamentals), including **CUET PG PYQs** where available.

☒ **Topic: Computer Arithmetic – Division Algorithms**

- ☒ Total: 30 MCQs
 - ☒ Correct answers are marked
 - ☒ CUET PG Previous Year Questions are tagged as **(PYQ)**
-

1. Which of the following is used in binary division?

- a) XOR operation
 - ☒ b) Repeated subtraction and shifting
 - c) Multiplexers
 - d) Parity generator
-

2. What is the quotient when 1010 is divided by 10 in binary?

- ☒ a) 101
 - b) 100
 - c) 10
 - d) 110
-

3. Binary division is similar to:

- a) Hex division
 - b) Gray code operation
 - ☒ c) Long division in decimal
 - d) Booth's algorithm
-

4. Restoring division algorithm uses:

- a) Addition
 - ☒ b) Subtraction and restoration
 - c) AND operation
 - d) OR operation
-

5. In restoring division, if remainder is negative:

- a) Proceed normally
 - ☒ b) Restore previous value
 - c) Shift right
 - d) Shift left
-

6. Which algorithm is more efficient for hardware division?

- ☒ a) Non-restoring division
 - b) Booth's algorithm
 - c) Wallace tree
 - d) Floating-point multiplication
-

7. In non-restoring division, when remainder is negative:

- a) Add divisor
 - ☒ b) Add divisor and set quotient bit to 0
 - c) Subtract again
 - d) Set overflow flag
-

8. In restoring division, what is done after subtraction?

- a) Right shift quotient
 - b) XOR with divisor
 - ☒ c) Check sign of remainder
 - d) Add quotient
-

9. Binary division: $1100 \div 10 = ?$

- a) 111
 - ☒ b) 110
 - c) 100
 - d) 101
-

10. The initial remainder in division is:

- ☒ a) Dividend
 - b) Divisor
 - c) Quotient
 - d) Zero
-

11. Division by shifting is possible only when divisor is:

- a) Odd
 - ☒ b) Power of 2
 - c) Even
 - d) Prime
-

12. 2's complement is used in division for:

- a) Multiplication
 - b) Parity checking
 - ☒ c) Signed number handling
 - d) Carry generation
-

13. (PYQ) CUET PG: Binary division of 10010 by 10 gives:

- ☒ a) 1001
 - b) 1010
 - c) 1011
 - d) 1100
-

14. What does shifting left in division operation do?

- a) Adds one
 - b) Multiplies
 - ☒ c) Prepares for subtraction
 - d) Changes sign
-

15. Which of the following does non-restoring division improve?

- ☒ a) Speed and performance
 - b) Accuracy
 - c) Clock frequency
 - d) Power usage
-

16. Overflow in binary division occurs when:

- a) Quotient is 0
 - ☒ b) Divisor is 0
 - c) Dividend is 0
 - d) Quotient is negative
-

17. Division algorithm terminates when:

- a) Remainder is 0
 - ☒ b) All bits are processed
 - c) Quotient is negative
 - d) MSB = LSB
-

18. Binary division: $1111 \div 11 = ?$

- a) 101
 - ☒ b) 101
 - c) 100
 - d) 110
-

19. Division of 16-bit numbers requires:

- a) 8-bit adder
 - b) 16-bit subtractor
 - ☒ c) 32-bit register (for quotient + remainder)
 - d) None of these
-

20. Quotient is stored in:

- ☒ a) Quotient register
 - b) ALU
 - c) Remainder register
 - d) Control unit
-

21. Signed division result sign rule:

- a) Positive always
 - ☒ b) Positive if signs are same
 - c) Negative if dividend is negative
 - d) Depends on LSB
-

22. In restoring division, what's the quotient bit when remainder ≥ 0 after subtraction?

- ☒ a) 1
 - b) 0
 - c) Unchanged
 - d) Undefined
-

23. The key difference in non-restoring division:

- a) No subtraction
 - b) No shift
 - ☒ c) No restoration step
 - d) Uses 1's complement
-

24. The hardware complexity of non-restoring division is:

- a) High
 - ☒ b) Lower than restoring
 - c) Very high
 - d) Same as multiplication
-

25. In fixed-point division, the fractional part is handled using:

- ☒ a) Shifts and normalization
 - b) Decimal conversion
 - c) Parity check
 - d) Sign extension
-

26. 0 divided by any number (non-zero) is:

- ☒ a) 0
 - b) Undefined
 - c) Error
 - d) Overflow
-

27. Division by zero is:

- a) Zero
 - ☒ b) Undefined
 - c) Negative
 - d) Signed overflow
-

28. Which signal is raised during divide-by-zero?

- a) ACK
 - b) RESET
 - ☒ c) Exception
 - d) HALT
-

29. Restoring algorithm is less efficient because:

- a) Incorrect results
 - ☒ b) Extra restoration step
 - c) Slow memory
 - d) No sign support
-

30. The main function of the ALU in division is:

- a) Input/output control
 - ☒ b) Subtract and shift
 - c) Encode data
 - d) Jump instruction
-

Here are **30 Multiple Choice Questions (MCQs)** on the topic "**Digital Computers**", including relevant **CUET PG PYQs** wherever applicable.

Topic: Digital Computers

- ☒ Total: 30 MCQs
 - ☒ Correct answers are marked
 - ☒ CUET PG Previous Year Questions are tagged as (PYQ)
-

1. A digital computer represents data using:

- a) Analog voltages
 - b) Continuous signals
 - ☒ c) **Discrete binary values**
 - d) Frequency waves
-

2. The heart of any digital computer is the:

- ☒ a) **Central Processing Unit (CPU)**
 - b) Monitor
 - c) RAM
 - d) ROM
-

3. Which of the following is not a characteristic of digital computers?

- a) High speed
 - ☒ b) **Continuous data representation**
 - c) Accuracy
 - d) Programmability
-

4. The unit that interprets and executes instructions is:

- a) ALU
 - ☒ b) **Control Unit**
 - c) Memory Unit
 - d) I/O Unit
-

5. Digital computers use the number system:

- a) Octal
 - b) Decimal
 - ☒ c) **Binary**
 - d) Hexadecimal
-

6. The ALU performs:

- a) Only arithmetic
 - b) Only logical
 - ☒ c) **Arithmetic and logical operations**
 - d) Memory operations
-

7. Memory used to store data permanently is:

- a) RAM
 - ☒ b) **ROM**
 - c) Cache
 - d) Register
-

8. In a digital computer, input devices convert:

- a) Binary to decimal
 - b) Logic to analog
 - ☒ c) **Analog signals to digital signals**
 - d) Text to audio
-

9. Which language is directly understood by the computer?

- a) High-level language
 - ☒ b) **Machine language**
 - c) Assembly language
 - d) C++
-

10. The component responsible for arithmetic operations is:

- ☒ a) **ALU**
 - b) Control Unit
 - c) ROM
 - d) CPU
-

11. What is the smallest unit of data in digital computers?

- a) Byte
 - ☒ b) Bit
 - c) Nibble
 - d) Word
-

12. The temporary memory used by CPU is:

- ☒ a) RAM
 - b) ROM
 - c) SSD
 - d) DVD
-

13. The binary value 1101 is equivalent to decimal:

- a) 11
 - ☒ b) 13
 - c) 15
 - d) 14
-

14. Digital computers differ from analog computers in terms of:

- a) Speed
 - ☒ b) Data representation
 - c) Size
 - d) Electricity
-

15. What stores the booting program of a computer?

- a) RAM
 - ☒ b) ROM
 - c) HDD
 - d) ALU
-

16. (PYQ) CUET PG: The basic unit of digital data is:

- a) Byte
 - ☒ b) Bit
 - c) Word
 - d) Instruction
-

17. The primary memory of a digital computer is:

- a) ROM
 - ☒ b) RAM
 - c) Flash
 - d) Hard disk
-

18. Which of these is not a digital computing device?

- ☒ a) Thermometer with mercury
 - b) Digital watch
 - c) Calculator
 - d) Laptop
-

19. The speed of a digital computer is measured in:

- a) Seconds
 - b) Volts
 - ☒ c) Hertz (Hz)
 - d) Amps
-

20. What part of CPU stores instructions temporarily during processing?

- a) ALU
 - ☒ b) Registers
 - c) Cache
 - d) I/O
-

21. Which computer generation used ICs (Integrated Circuits)?

- a) First
 - b) Second
 - ☒ c) **Third**
 - d) Fourth
-

22. (PYQ) CUET PG: Which component performs decision-making tasks?

- a) Control Unit
 - ☒ b) **ALU**
 - c) Monitor
 - d) Keyboard
-

23. Which part coordinates all parts of the computer?

- a) ALU
 - ☒ b) **Control Unit**
 - c) Monitor
 - d) CPU
-

24. Which of the following is not an output device?

- a) Monitor
 - ☒ b) **Mouse**
 - c) Printer
 - d) Speaker
-

25. The program that starts automatically when the computer is turned on:

- ☒ a) **BIOS**
 - b) Compiler
 - c) Assembler
 - d) Interpreter
-

26. Which number system is most commonly used in computers?

- a) Octal
 - ☒ b) **Binary**
 - c) Hexadecimal
 - d) Ternary
-

27. Decimal number 15 in binary is:

- ☒ a) **1111**
 - b) 1010
 - c) 1001
 - d) 1100
-

28. What does a digital computer use to process data?

- a) Continuous signals
 - ☒ b) **Logical and arithmetic operations**
 - c) Analog dials
 - d) Sound waves
-

29. A device that converts digital signals into sound:

- a) Keyboard
 - b) Monitor
 - ☒ c) **Speaker**
 - d) Mouse
-

30. Which memory is both readable and writable?

- a) ROM
 - b) PROM
 - ☒ c) **RAM**
 - d) CD-ROM
-

Here are **30 Multiple Choice Questions (MCQs)** on **Logic Gates**, including **CUET PG Previous Year Questions (PYQs)** wherever applicable.

☐ **Topic: Logic Gates**

☐ Total: 30 MCQs

☐ Correct answers marked

☐ PYQs marked as **(PYQ CUET PG)**

1. The basic logic gates are:

☐ **a) AND, OR, NOT**

b) NAND, NOR, XOR

c) AND, XOR, XNOR

d) NOR, NOT, XOR

2. Which logic gate gives an output of 1 only when both inputs are 1?

a) OR

☐ **b) AND**

c) NOT

d) NAND

3. Which logic gate gives an output of 0 when both inputs are 1?

☐ **a) NAND**

b) AND

c) OR

d) XOR

4. The symbol for the OR gate is:

a) •

b) +

☐ **c) ≥ 1**

d) $\oplus$

5. The NOT gate is also called a:

- a) Buffer
 - ☒ b) Inverter
 - c) Selector
 - d) Decoder
-

6. The output of an XOR gate is 1 when:

- ☒ a) Inputs are different
 - b) Inputs are the same
 - c) All inputs are 1
 - d) All inputs are 0
-

7. Which gate is known as a universal gate?

- ☒ a) NAND
 - b) AND
 - c) OR
 - d) XOR
-

8. A gate that gives a high output when all inputs are low is:

- a) OR
 - b) AND
 - ☒ c) NOR
 - d) NAND
-

9. The output of a NOT gate is:

- ☒ a) Complement of input
 - b) Same as input
 - c) Input AND 1
 - d) Always 0
-

10. Which gate is the combination of an AND gate followed by a NOT gate?

- ☐ a) **NAND**
 - b) NOR
 - c) XOR
 - d) XNOR
-

11. Which logic gate outputs 1 only when both inputs are different?

- a) XNOR
 - ☐ b) **XOR**
 - c) OR
 - d) NOR
-

12. What is the output of a NOR gate when both inputs are 0?

- ☐ a) **1**
 - b) 0
 - c) Undefined
 - d) Depends on the system
-

13. (PYQ CUET PG) Which of the following gates can be used to construct any logic circuit?

- ☐ a) **NAND and NOR**
 - b) AND and OR
 - c) XOR and XNOR
 - d) AND and NOT
-

14. What is the output of an AND gate if both inputs are 0?

- ☐ a) **0**
 - b) 1
 - c) Undefined
 - d) Depends on the power supply
-

15. A NAND gate is a combination of:

- a) OR + NOT
 - ☒ b) AND + NOT
 - c) XOR + NOT
 - d) NOR + NOT
-

16. The truth table of an OR gate shows output 1 if:

- ☒ a) At least one input is 1
 - b) All inputs are 0
 - c) All inputs are 1
 - d) Inputs are different
-

17. An XNOR gate gives high output when:

- a) Inputs differ
 - ☒ b) Inputs are the same
 - c) One input is 0
 - d) One input is 1
-

18. Which gate's symbol looks like a triangle with a small circle?

- ☒ a) NOT
 - b) AND
 - c) OR
 - d) NAND
-

19. If inputs of a NOR gate are $A = 0$, $B = 1$, output is:

- a) 1
 - ☒ b) 0
 - c) Undefined
 - d) Can't be determined
-

20. Which gate is equivalent to an OR gate followed by NOT?

- ☒ a) NOR
 - b) NAND
 - c) XOR
 - d) AND
-

21. The minimum number of NAND gates required to implement a NOT gate is:

- ☒ a) 1
 - b) 2
 - c) 3
 - d) 0
-

22. Which logic gate has an output expression $A \cdot \bar{B} + \bar{A} \cdot B$?

- a) NOR
 - b) XNOR
 - ☒ c) XOR
 - d) NAND
-

23. A combination of XOR and NOT gate gives:

- ☒ a) XNOR
 - b) OR
 - c) NOR
 - d) NAND
-

24. Which logic gate does not exist as a basic gate?

- a) AND
 - ☒ b) XNOR
 - c) NOT
 - d) OR
-

25. The output of an AND gate is 1 if:

- ☐ a) All inputs are 1
 - b) One input is 0
 - c) One input is 1
 - d) Inputs differ
-

26. (PYQ CUET PG) Which gate gives 0 output when inputs are same?

- a) XNOR
 - ☐ b) XOR
 - c) OR
 - d) AND
-

27. In digital electronics, the gate that acts as a basic decision maker is:

- ☐ a) AND
 - b) NOT
 - c) NOR
 - d) XNOR
-

28. NAND and NOR gates are called universal gates because:

- ☐ a) They can implement all basic gates
 - b) Used in all circuits
 - c) Easy to construct
 - d) Only used in computers
-

29. A buffer is a logic gate with:

- ☐ a) No inversion
 - b) Two outputs
 - c) No input
 - d) Only output
-

30. The output of a NOT gate when input is 1:

- a) 1
 - ☒ b) 0
 - c) Undefined
 - d) Depends on circuit
-

Here are **30 MCQs on “Boolean Algebra”** for **CUET PG MCA** preparation, including **CUET PG Previous Year Questions (PYQs)** where applicable.

☐ **Topic: Boolean Algebra**

☐ Total: 30 MCQs

☐ Correct answers are marked

☐ **PYQ = Previous Year CUET PG Question**

1. Boolean algebra is primarily used in:

- a) Numerical analysis
 - ☒ b) Digital circuit design
 - c) Statistics
 - d) Calculus
-

2. Boolean variables can have:

- a) 0 to 9 values
 - b) Infinite values
 - ☒ c) Two values (0 and 1)
 - d) Only 1 value
-

3. (PYQ CUET PG) The identity law in Boolean algebra is:

- ☒ a) $A + 0 = A$ and $A \cdot 1 = A$
 - b) $A + 1 = A$ and $A \cdot 0 = A$
 - c) $A + A = A$ and $A \cdot A = 1$
 - d) $A + A' = 1$ and $A \cdot A' = A$
-

4. The complement of $A + AB$ is:

- a) $A' + B$
 - ☒ b) $A'B'$
 - c) AB'
 - d) $A' + B'$
-

5. The expression $A + A'B$ simplifies to:

- a) AB
 - ☒ b) $A + B$
 - c) $A' + B$
 - d) $A + A'$
-

6. The absorption law is:

- a) $A + 0 = A$
 - ☒ b) $A + AB = A$
 - c) $A + A = A$
 - d) $A \cdot A = A$
-

7. The complement law is:

- ☒ a) $A + A' = 1$ and $A \cdot A' = 0$
 - b) $A + A = A$ and $A \cdot A = A$
 - c) $A + 0 = A$
 - d) $A \cdot 1 = A$
-

8. $A + AB = ?$

- a) $A \cdot B$
 - ☒ b) A
 - c) B
 - d) $A + B$
-

9. (PYQ CUET PG) Which of the following is the involution law?

- a) $A + A = A$
 - b) $A \cdot A = A$
 - ☐ c) $(A')' = A$
 - d) $A + A' = 1$
-

10. The dual of $A + 0 = A$ is:

- ☐ a) $A \cdot 1 = A$
 - b) $A + 1 = A$
 - c) $A \cdot 0 = 0$
 - d) $A + A = A$
-

11. De Morgan's first law:

- ☐ a) $(A + B)' = A' \cdot B'$
 - b) $(A \cdot B)' = A + B$
 - c) $A + A' = 1$
 - d) $A \cdot A' = A$
-

12. The dual of $A \cdot (B + C) = A \cdot B + A \cdot C$ is:

- a) $A + (B \cdot C) = A + B \cdot A + C$
 - ☐ b) $A + (B \cdot C) = (A + B) \cdot (A + C)$
 - c) $A + (B \cdot C) = (A \cdot B) + (A \cdot C)$
 - d) $A + B \cdot C = A \cdot B + C$
-

13. Which of the following is a valid Boolean identity?

- ☐ a) $A + A = A$
 - b) $A + A = 1$
 - c) $A \cdot A = 0$
 - d) $A \cdot A = 1$
-

14. Which of the following is a redundant term in $A + AB$?

- ☐ a) AB
 - b) A
 - c) None
 - d) $A + B$
-

15. The result of $A + A'B$ is:

- a) AB
 - b) $A' + B$
 - ☐ c) $A + B$
 - d) $A \cdot B$
-

16. A Boolean expression that equals 0 for all inputs is called:

- ☐ a) Contradiction
 - b) Tautology
 - c) Identity
 - d) Equivalence
-

17. What is $A + A'$ equal to?

- a) 0
 - b) A
 - ☐ c) 1
 - d) A'
-

18. The distributive law is:

- ☐ a) $A \cdot (B + C) = A \cdot B + A \cdot C$
 - b) $A + (B \cdot C) = A + B + C$
 - c) $A \cdot B + A = A$
 - d) $A + B = B + A$
-

19. A logic circuit using Boolean equation $A + B'C$ means:

- a) A OR B AND C
 - ☒ b) A OR (NOT B AND C)
 - c) A AND (B OR C)
 - d) A AND B OR C
-

20. (PYQ CUET PG) Which of the following is correct by De Morgan's law?

- a) $(AB)' = A'B'$
 - ☒ b) $(A + B)' = A' \cdot B'$
 - c) $(A + B)' = A + B$
 - d) $(AB)' = A + B$
-

21. The simplification of $A(A + B)$ is:

- a) AB
 - b) $A + AB$
 - ☒ c) A
 - d) $A + B$
-

22. The expression $(A + B)(A + C)$ simplifies to:

- ☒ a) $A + BC$
 - b) $AB + AC$
 - c) $A + B + C$
 - d) $A + B + AC$
-

23. What is the value of $A \cdot A'$?

- a) 1
 - ☒ b) 0
 - c) A
 - d) A'
-

24. Which law proves $A + A = A$?

- a) Complement
 - ☒ b) Idempotent
 - c) Associative
 - d) Distributive
-

25. What is the result of $(A + B) \cdot (A + C)$?

- a) $A + BC$
 - b) $AB + AC$
 - ☒ c) $A + BC$
 - d) $A + B + C$
-

26. Which of these is not a Boolean law?

- a) Identity
 - ☒ b) Quadratic law
 - c) Distributive
 - d) Complement
-

27. Which Boolean property is used in $A \cdot (A + B) = A$?

- ☒ a) Absorption
 - b) Distributive
 - c) Demorgan
 - d) Associative
-

28. What is the result of $(A + B)(A + C)$ using Boolean simplification?

- a) $A + B + C$
 - ☒ b) $A + BC$
 - c) $AB + AC$
 - d) $A \cdot (B + C)$
-

29. Which expression is equal to 1 for all values of A?

- a) $A \cdot A$
 - ☒ b) $A + A'$
 - c) $A \cdot A'$
 - d) A
-

30. The Boolean equivalent of the expression $(A \cdot B) + (A \cdot B')$ is:

- a) AB
 - b) AB'
 - ☒ c) A
 - d) B
-

Here are **30 MCQs on “Map Simplifications (Karnaugh Map/K-Map Simplification)”** for **CUET PG MCA** preparation, including **CUET PG Previous Year Questions (PYQs)** where applicable.

☒ **Topic: Map Simplifications (K-Map)**

☒ Total: 30 MCQs

☒ Correct answers are marked

☒ PYQ = CUET PG Previous Year Question

1. The Karnaugh Map is used for:

- a) Data transmission
 - b) Hardware design
 - ☒ c) **Boolean expression simplification**
 - d) Programming logic
-

2. K-map is a diagrammatic method used for:

- ☒ a) **Minimizing Boolean functions**
 - b) Storing binary data
 - c) Compiling code
 - d) Decoding inputs
-

3. A 2-variable K-map has:

- a) 2 cells
 - ☒ **b) 4 cells**
 - c) 8 cells
 - d) 16 cells
-

4. A 3-variable K-map has:

- a) 6 cells
 - ☒ **b) 8 cells**
 - c) 12 cells
 - d) 16 cells
-

5. A 4-variable K-map has:

- a) 8 cells
 - b) 12 cells
 - ☒ **c) 16 cells**
 - d) 24 cells
-

6. A 5-variable K-map has:

- a) 16 cells
 - b) 24 cells
 - ☒ **c) 32 cells**
 - d) 64 cells
-

7. (PYQ CUET PG) Adjacent 1's in K-map are grouped in:

- a) Even numbers
 - ☒ **b) Powers of 2**
 - c) Multiples of 3
 - d) Prime numbers
-

8. The Gray code is used in K-maps to ensure:

- ☐ a) Only one-bit change between adjacent cells
 - b) Binary order is maintained
 - c) Maximum 1s are together
 - d) Truth table is reversed
-

9. In K-map simplification, a group of 8 adjacent 1's represents:

- ☐ a) 3-variable elimination
 - b) 3-variable inclusion
 - c) Single variable
 - d) No simplification
-

10. Don't care conditions in K-map are denoted by:

- ☐ a) X
 - b) D
 - c) 0
 - d) 1
-

11. Don't care terms can be used to:

- a) Increase minterms
 - ☐ b) Simplify the function further
 - c) Complicate the logic
 - d) Convert to maxterms
-

12. A group of 4 adjacent 1s in a 3-variable K-map simplifies to:

- ☐ a) A 1-variable expression
 - b) 3-variable expression
 - c) No simplification
 - d) Constant value
-

13. Minimum number of adjacent squares required to eliminate two variables is:

- a) 2
 - b) 4
 - ☒ c) 4
 - d) 8
-

14. The output of the function with K-map grouping 4 adjacent 1's results in:

- a) Single literal
 - ☒ b) 2-variable term
 - c) Unchanged expression
 - d) Don't care condition
-

15. (PYQ CUET PG) What is the simplified form of the expression using K-map for minterms (1,3,5,7)?

- a) A
 - b) B
 - ☒ c) $A + B$
 - d) $A \cdot B$
-

16. Grouping two 1s in K-map eliminates:

- a) 2 variables
 - ☒ b) 1 variable
 - c) No variable
 - d) All variables
-

17. Which of the following group sizes are invalid in K-map?

- a) 1
 - b) 2
 - ☒ c) 3
 - d) 4
-

18. Essential prime implicants are:

- ☒ a) Must be included in the simplified function
 - b) Can be ignored
 - c) Used only in SOP
 - d) Formed from zeros
-

19. Prime implicants are:

- ☒ a) All possible groupings of 1's in K-map
 - b) Groupings of don't cares only
 - c) Formed using minterms of 0
 - d) Variables used once
-

20. How many prime implicants are possible from a single 1 in K-map?

- ☒ a) One
 - b) Two
 - c) Infinite
 - d) Zero
-

21. The largest possible group in a 4-variable K-map is:

- a) 4
 - ☒ b) 8
 - c) 16
 - d) 32
-

22. If a 4-variable K-map has minterms (0,1,2,3), then simplified form is:

- ☒ a) $A'B'$
 - b) AB
 - c) $A + B$
 - d) $A' + B'$
-

23. A loop in K-map must contain:

☐ a) Powers of 2 number of 1s

b) Any number of terms

c) Only 2s

d) Prime numbers

24. The main advantage of using K-map over algebraic methods:

a) Converts to decimal

b) Solves for 0s

☐ c) Easier and visual simplification

d) Slower but accurate

25. If you group all 1s in a 3-variable K-map, output is:

☐ a) 1

b) 0

c) A

d) B

26. A group that contains don't care and 1s is valid if:

a) All don't cares are used

☐ b) It helps simplification

c) Don't care is ignored

d) Grouping forms 3

27. The wrap-around grouping in a K-map refers to:

a) Linear arrangement

☐ b) Edge adjacency in map

c) Circular gates

d) Matrix flipping

28. Simplification of SOP using K-map involves:

- ☒ a) Grouping 1s
 - b) Grouping 0s
 - c) XOR operation
 - d) NOR logic
-

29. POS simplification in K-map involves:

- a) Grouping 1s
 - ☒ b) Grouping 0s
 - c) Grouping X
 - d) Complementing SOP
-

30. Simplified form of minterms (2,3,6,7) in 3-variable K-map is:

- ☒ a) B
 - b) A
 - c) A + C
 - d) AB
-

Here are **30 MCQs on "Combinational Circuits"** for **CUET PG MCA** preparation, including **CUET PG Previous Year Questions (PYQs)** where available.

☒ **Topic: Combinational Circuits**

- ☒ Total: 30 MCQs
 - ☒ Correct answers are marked
 - ☒ PYQ = CUET PG Previous Year Question
-

1. A combinational circuit consists of:

- a) Memory elements
 - ☒ b) Logic gates only
 - c) Flip-flops and registers
 - d) Clock signals
-

2. A half-adder adds:

- a) 2-bit binary numbers
 - ☒ **b) Two single bits**
 - c) 4-bit numbers
 - d) Signed integers
-

3. The number of outputs in a full adder is:

- a) 1
 - ☒ **b) 2**
 - c) 3
 - d) 4
-

4. A full adder adds:

- a) Only two bits
 - ☒ **b) Two bits and a carry**
 - c) Three 4-bit numbers
 - d) Decimal digits
-

5. A multiplexer is also known as:

- ☒ **a) Data selector**
 - b) Decoder
 - c) Encoder
 - d) Demultiplexer
-

6. The output of a 2x1 multiplexer is:

- a) 0
 - ☒ **b) One of the two inputs**
 - c) XOR of inputs
 - d) Always 1
-

7. Which of the following is not a combinational circuit?

- ☒ a) Flip-flop
 - b) Decoder
 - c) Encoder
 - d) Multiplexer
-

8. A decoder converts:

- ☒ a) Binary information to 2^n outputs
 - b) 2^n outputs to binary
 - c) Decimal to binary
 - d) Octal to hexadecimal
-

9. A 4-to-16 decoder has how many input and output lines?

- a) 2 inputs, 8 outputs
 - ☒ b) 4 inputs, 16 outputs
 - c) 8 inputs, 4 outputs
 - d) 16 inputs, 4 outputs
-

10. The function of an encoder is to:

- a) Decode signals
 - ☒ b) Convert 2^n inputs to binary outputs
 - c) Multiply numbers
 - d) Add binary digits
-

11. A priority encoder resolves:

- a) Gate delay
 - ☒ b) Conflicts among multiple active inputs
 - c) Overflow conditions
 - d) Memory access
-

12. A comparator compares:

- a) Binary with decimal
 - b) Two 4-bit numbers only
 - ☒ c) **Two binary numbers**
 - d) Signed integers
-

13. A 1-bit comparator outputs:

- a) Only 0
 - b) Only 1
 - ☒ c) **Greater, Less, Equal**
 - d) XOR result
-

14. A Demultiplexer is used to:

- a) Select inputs
 - ☒ b) **Route one input to multiple outputs**
 - c) Add binary numbers
 - d) Store data
-

15. Number of select lines for 8-to-1 MUX:

- a) 2
 - ☒ b) **3**
 - c) 4
 - d) 1
-

16. (PYQ CUET PG) A Half-Adder cannot:

- a) Add bits
 - ☒ b) **Handle carry-in**
 - c) Output sum
 - d) Use XOR gate
-

17. A Full-Adder can be built using:

- a) One XOR and one AND gate
 - ☒ b) Two half-adders and one OR gate
 - c) Only OR gates
 - d) Two NAND gates
-

18. A 2-to-4 decoder requires how many input lines?

- ☒ a) 2
 - b) 4
 - c) 3
 - d) 1
-

19. Which logic gate is essential for a half-adder?

- a) OR
 - b) NOR
 - ☒ c) XOR
 - d) XNOR
-

20. The Boolean expression for sum output of a half-adder is:

- ☒ a) $A \oplus B$
 - b) $A + B$
 - c) $A \cdot B$
 - d) $A + AB$
-

21. The Boolean expression for carry output of a half-adder is:

- a) $A \oplus B$
 - b) $A + B$
 - ☒ c) $A \cdot B$
 - d) $A \oplus \bar{B}$
-

22. A 4-to-1 multiplexer has how many selection lines?

- a) 1
 - ☒ b) 2
 - c) 4
 - d) 3
-

23. What is the output of a 1-to-4 demux when input is 1 and select lines are "10"?

- a) Line 0
 - b) Line 1
 - ☒ c) Line 2
 - d) Line 3
-

24. Which combinational circuit is used in ALU for binary addition?

- a) Decoder
 - b) MUX
 - ☒ c) Adder
 - d) Demux
-

25. A combinational circuit outputs depend on:

- ☒ a) Current inputs only
 - b) Previous inputs
 - c) Clock inputs
 - d) Storage values
-

26. The delay in combinational circuits depends on:

- a) RAM
 - ☒ b) Number of logic levels
 - c) Flip-flops used
 - d) Input voltage
-

27. The number of outputs of a 3-to-8 decoder is:

- a) 3
 - ☒ b) 8
 - c) 24
 - d) 4
-

28. A half-subtractor is used to:

- a) Add decimal digits
 - ☒ b) Subtract 2 bits
 - c) Multiply numbers
 - d) Detect overflow
-

29. Full-subtractor has how many inputs?

- a) 2
 - ☒ b) 3
 - c) 4
 - d) 1
-

30. The difference output of a half-subtractor is:

- ☒ a) $A \oplus B$
 - b) $A \cdot B$
 - c) $A + B$
 - d) $A + AB$
-

☒ **Topic: Flip-Flops**

☒ Total: 30 MCQs

☒ Correct answers are marked

☒ PYQ = CUET PG Previous Year Question

1. Flip-Flops are:

- a) Combinational circuits
 - ☒ b) **Sequential circuits**
 - c) Arithmetic circuits
 - d) Linear circuits
-

2. A flip-flop can store:

- a) 2 bits
 - ☒ b) **1 bit**
 - c) 4 bits
 - d) 8 bits
-

3. The basic memory element in sequential logic is:

- ☒ a) **Flip-Flop**
 - b) Counter
 - c) Register
 - d) Gate
-

4. SR flip-flop gives invalid output when:

- a) $S = 0, R = 1$
 - ☒ b) **$S = 1, R = 1$**
 - c) $S = 1, R = 0$
 - d) $S = 0, R = 0$
-

5. The JK flip-flop overcomes the invalid condition of:

- ☒ a) **SR flip-flop**
 - b) D flip-flop
 - c) T flip-flop
 - d) Master-slave flip-flop
-

6. A T flip-flop toggles when:

- ☒ a) $T = 1$
 - b) $T = 0$
 - c) Clock is 0
 - d) $S = R = 1$
-

7. D flip-flop eliminates the problem of:

- a) Race-around
 - ☒ b) Invalid inputs of SR flip-flop
 - c) Clock glitches
 - d) Data delay
-

8. The characteristic equation of a D flip-flop is:

- a) $Q(t+1) = Q$
 - ☒ b) $Q(t+1) = D$
 - c) $Q(t+1) = D'$
 - d) $Q(t+1) = D \oplus Q$
-

9. The output of a flip-flop depends on:

- a) Current input only
 - ☒ b) Current input and previous state
 - c) Clock speed only
 - d) Number of gates
-

10. A flip-flop changes state:

- ☒ a) On clock pulse
 - b) On power-on
 - c) Randomly
 - d) Asynchronously only
-

11. A JK flip-flop toggles when:

- ☒ a) $J = 1, K = 1$
 - b) $J = 0, K = 0$
 - c) $J = 1, K = 0$
 - d) $J = 0, K = 1$
-

12. A master-slave flip-flop is made using:

- a) One JK flip-flop
 - ☒ b) **Two JK flip-flops**
 - c) One SR and one JK
 - d) One D and one T
-

13. In a T flip-flop, when $T=1$, output:

- a) Remains same
 - ☒ b) **Toggles**
 - c) Becomes 0
 - d) Becomes 1
-

14. Flip-flops are used in:

- ☒ a) **Counters**
 - b) Adders
 - c) Multiplexers
 - d) Decoders
-

15. The input combination $S=R=1$ in SR flip-flop causes:

- a) Set
 - b) Reset
 - ☒ c) **Invalid state**
 - d) Toggle
-

16. (PYQ CUET PG) Which flip-flop has only one input?

- a) SR
 - ☒ b) D
 - c) JK
 - d) Master-Slave JK
-

17. The number of states in 3 flip-flops is:

- a) 6
 - ☒ b) 8
 - c) 2
 - d) 16
-

18. Edge-triggered flip-flops are sensitive to:

- a) Level
 - ☒ b) Clock edge
 - c) Voltage
 - d) Current
-

19. A clock signal is used to:

- ☒ a) Synchronize flip-flop operation
 - b) Store inputs
 - c) Detect errors
 - d) Delay outputs
-

20. In a JK flip-flop, $J = K = 0$ gives:

- ☒ a) No change
 - b) Reset
 - c) Set
 - d) Toggle
-

21. The race-around condition occurs in:

- ☒ a) JK flip-flop
 - b) SR flip-flop
 - c) T flip-flop
 - d) D flip-flop
-

22. Which flip-flop has no invalid state?

- ☒ a) D
 - b) SR
 - c) JK
 - d) T
-

23. The toggle flip-flop is also known as:

- ☒ a) T flip-flop
 - b) D flip-flop
 - c) RS flip-flop
 - d) Clock divider
-

24. Which flip-flop stores input directly on clock edge?

- a) T flip-flop
 - b) SR flip-flop
 - ☒ c) D flip-flop
 - d) JK flip-flop
-

25. In D flip-flop, if D = 1 and clock is triggered, then Q becomes:

- a) 0
 - ☒ b) 1
 - c) Undefined
 - d) Complement
-

26. SR flip-flop truth table has how many valid states?

- a) 2
 - b) 4
 - ☐ c) 3
 - d) 1
-

27. JK flip-flop becomes D flip-flop if:

- a) $J = 0$
 - ☐ b) $K = J'$
 - c) $K = 0$
 - d) $J = K$
-

28. Flip-flops can be used to build:

- a) Adders
 - ☐ b) Registers
 - c) Encoders
 - d) Multiplexers
-

29. A flip-flop is a:

- ☐ a) Bistable device
 - b) Monostable device
 - c) Multistable device
 - d) Astable device
-

30. Number of flip-flops needed to store 16 states:

- a) 3
 - ☐ b) 4
 - c) 2
 - d) 5
-

Here are 30 MCQs on “Sequential Circuits” for CUET PG MCA preparation, including CUET PG Previous Year Questions (PYQs) wherever applicable.

☐ **Topic: Sequential Circuits**

☐ Total: 30 MCQs

☐ Correct answers are marked

☐ PYQ = CUET PG Previous Year Question

1. What differentiates sequential circuits from combinational circuits?

- a) They have no inputs
 - ☐ **b) They have memory elements**
 - c) They are analog
 - d) They are non-digital
-

2. Sequential circuits use which type of logic devices?

- a) Adders
 - b) Multiplexers
 - ☐ **c) Flip-flops**
 - d) Encoders
-

3. Which of the following is a sequential circuit?

- a) Decoder
 - ☐ **b) Counter**
 - c) Adder
 - d) Multiplexer
-

4. The output of a sequential circuit depends on:

- a) Present input only
 - b) Clock only
 - ☐ **c) Present input and past output (state)**
 - d) Power supply
-

5. What is the basic building block of sequential circuits?

- a) Logic gates
 - ☒ b) Flip-Flops
 - c) Adders
 - d) ROM
-

6. A sequential circuit with no clock is called:

- ☒ a) Asynchronous
 - b) Synchronous
 - c) Combinational
 - d) Ripple
-

7. In a synchronous circuit, all flip-flops are triggered by:

- a) Independent clocks
 - ☒ b) Same clock
 - c) Flip-flop output
 - d) Clock divider
-

8. An example of a synchronous sequential circuit is:

- a) Multiplexer
 - ☒ b) Shift register
 - c) Encoder
 - d) Logic gate
-

9. The state of a sequential circuit is stored in:

- ☒ a) Flip-flops
 - b) Multiplexers
 - c) RAM
 - d) Cache
-

10. A sequential circuit uses 3 flip-flops. How many states can it represent?

- a) 4
 - ☒ b) 8
 - c) 6
 - d) 12
-

11. A counter is a:

- a) Combinational device
 - ☒ b) **Sequential device**
 - c) Decoder
 - d) Multiplexer
-

12. (PYQ CUET PG) Mealy and Moore machines are types of:

- a) Combinational circuits
 - ☒ b) **Sequential machines**
 - c) RAM devices
 - d) ALUs
-

13. In Moore machine, output depends on:

- ☒ a) **Current state only**
 - b) Input only
 - c) Previous state
 - d) Clock pulse
-

14. In Mealy machine, output depends on:

- a) Present state only
 - ☒ b) **Present state and input**
 - c) Flip-flop type
 - d) Reset line
-

15. A finite state machine (FSM) is used to model:

- a) Analog devices
 - ☒ b) Sequential logic behavior
 - c) ROM memory
 - d) CPU cache
-

16. What does the "state" in a sequential circuit represent?

- ☒ a) Stored past behavior
 - b) Input voltage
 - c) Clock frequency
 - d) Memory size
-

17. What is the minimum number of flip-flops required to build a MOD-6 counter?

- a) 2
 - ☒ b) 3
 - c) 4
 - d) 6
-

18. In a sequential circuit, feedback is provided from:

- a) Input
 - ☒ b) Output to input
 - c) Clock to data
 - d) Decoder
-

19. A clock pulse in sequential circuits:

- ☒ a) Synchronizes state changes
 - b) Increases delay
 - c) Deletes memory
 - d) Multiplies data
-

20. The speed of a sequential circuit is mainly limited by:

- a) Input size
 - ☒ b) Propagation delay of flip-flops
 - c) Power consumption
 - d) Instruction size
-

21. The output of an asynchronous sequential circuit may be unstable due to:

- ☒ a) Hazard/race conditions
 - b) Flip-flop count
 - c) High power
 - d) Constant inputs
-

22. A state diagram shows:

- a) Voltage levels
 - ☒ b) Transitions between states
 - c) Current values
 - d) Gate delays
-

23. A shift register is a type of:

- a) Decoder
 - ☒ b) Sequential circuit
 - c) Combinational circuit
 - d) MUX
-

24. What causes a transition between states in a synchronous sequential circuit?

- a) Data input
 - ☒ b) Clock signal
 - c) Reset
 - d) Voltage fluctuation
-

25. The minimum number of states in a MOD-3 counter is:

- a) 4
 - b) 1
 - ☒ c) 3
 - d) 5
-

26. A ring counter is a type of:

- ☒ a) Sequential circuit
 - b) Combinational circuit
 - c) Arithmetic circuit
 - d) Analog circuit
-

27. What type of sequential circuit outputs do not change immediately after input?

- ☒ a) Moore machine
 - b) Mealy machine
 - c) ROM-based
 - d) Clockless
-

28. Which sequential circuit reduces state count by eliminating redundant states?

- a) Ring counter
 - b) Synchronous counter
 - ☒ c) State minimization algorithm
 - d) Flip-flop
-

29. A MOD-8 counter requires:

- a) 2 flip-flops
 - ☒ b) 3 flip-flops
 - c) 4 flip-flops
 - d) 8 flip-flops
-

30. The difference between synchronous and asynchronous counters is:

- a) Number of states
 - ☐ b) **Clock triggering method**
 - c) Size
 - d) Voltage
-

Here are **30 MCQs on "Integrated Circuits (ICs)"** for **CUET PG MCA – Digital Fundamentals**, including **CUET PG previous year questions (PYQs)** where available.

☐ **Topic: Integrated Circuits (ICs)**

- ☐ Total Questions: 30
 - ☐ Correct answers are marked
 - ☐ PYQ = CUET PG Previous Year Question
-

1. What does IC stand for?

- a) Internal Circuit
 - b) Intelligent Circuit
 - ☐ c) **Integrated Circuit**
 - d) Interface Circuit
-

2. An Integrated Circuit is a:

- a) Single logic gate
 - b) Collection of wires
 - ☐ c) **Miniaturized electronic circuit on a semiconductor**
 - d) Group of flip-flops
-

3. ICs are primarily made using which material?

- ☐ a) **Silicon**
 - b) Copper
 - c) Germanium
 - d) Graphene
-

4. What is SSI in the context of ICs?

- ☒ a) **Small Scale Integration**
 - b) Super Speed Integration
 - c) Semi-Standard Integration
 - d) Static System Integration
-

5. Which type of IC has 1000 to 9999 gates per chip?

- a) SSI
 - b) MSI
 - ☒ c) **LSI**
 - d) VLSI
-

6. Which of the following best describes VLSI?

- ☒ a) **Very Large Scale Integration**
 - b) Variable Logic System Interface
 - c) Virtual Logic Signal Interference
 - d) Verified Logical Structured IC
-

7. VLSI circuits typically contain:

- a) <10 gates
 - ☒ b) **>10,000 gates**
 - c) Exactly 512 gates
 - d) No logic gates
-

8. Which of the following devices is a VLSI component?

- a) Full adder
 - b) AND gate
 - ☒ c) **Microprocessor**
 - d) Diode
-

9. MSI (Medium Scale Integration) ICs typically contain:

- ☒ a) 10 to 100 gates
 - b) 1 to 10 gates
 - c) 100 to 1000 gates
 - d) Over 10,000 gates
-

10. Which technology is used to manufacture ICs?

- ☒ a) Photolithography
 - b) Etching with acids
 - c) Wire bonding only
 - d) Soldering manually
-

11. The first commercially successful IC was developed in:

- a) 1945
 - b) 1955
 - ☒ c) 1959
 - d) 1969
-

12. (PYQ CUET PG) ICs are preferred in digital electronics mainly because:

- a) They use more wires
 - ☒ b) They are compact and reliable
 - c) They are heavy and expensive
 - d) They consume more power
-

13. What is an op-amp?

- a) A logic gate
 - b) Flip-flop circuit
 - ☒ c) Analog Integrated Circuit
 - d) Power supply
-

14. What are digital ICs mainly composed of?

- ☒ a) Logic gates
 - b) Inductors
 - c) Op-amps
 - d) Transformers
-

15. The packaging that holds the IC is known as:

- a) Board
 - ☒ b) Chip carrier
 - c) Logic frame
 - d) Bus unit
-

16. A dual in-line package (DIP) has:

- a) One row of pins
 - ☒ b) Two parallel rows of pins
 - c) No pins
 - d) Circular pin layout
-

17. IC fabrication technology is based on:

- a) Vacuum tubes
 - ☒ b) Planar technology
 - c) Coaxial layouts
 - d) Magnetic tape
-

18. ICs are classified based on:

- a) Color
 - ☒ b) Number of gates/components
 - c) Shape
 - d) Clock frequency
-

19. Logic families like TTL and CMOS are examples of:

☐ a) Digital IC technologies

b) Magnetic logic

c) Analog op-amps

d) Mechanical relays

20. Which logic family is known for low power consumption?

a) ECL

☐ b) CMOS

c) TTL

d) RTL

21. CMOS stands for:

a) Central Metal Oxide Semiconductor

☐ b) Complementary Metal Oxide Semiconductor

c) Common MOS

d) Coded MOS

22. ICs are connected to external circuits through:

a) Coils

☐ b) Pins

c) Screens

d) Switches

23. The size of components inside an IC is measured in:

a) Centimeters

☐ b) Microns or nanometers

c) Inches

d) Degrees

24. In TTL logic family, the basic logic gate is built using:

- ☒ a) **Bipolar junction transistors**
 - b) MOSFETs
 - c) Diodes
 - d) Relays
-

25. Which generation of ICs introduced microprocessors?

- a) 1st
 - b) 2nd
 - ☒ c) **3rd**
 - d) 4th
-

26. ICs are typically fabricated on:

- ☒ a) **Silicon wafers**
 - b) Metal boards
 - c) Plastic films
 - d) Glass plates
-

27. What does ULSI stand for?

- ☒ a) **Ultra Large Scale Integration**
 - b) Unified Logic System Integration
 - c) Ultra Logic System Interface
 - d) Universal Logic Signal Integration
-

28. Compared to discrete circuits, ICs:

- a) Use more power
 - b) Are slower
 - ☒ c) **Occupy less space**
 - d) Need cooling fans
-

29. Which is NOT a feature of integrated circuits?

- a) Compact
 - b) Low power
 - ☒ c) Easily repairable
 - d) Mass-producible
-

30. A System on Chip (SoC) is an example of:

- ☒ a) VLSI
 - b) MSI
 - c) SSI
 - d) Analog circuit
-

Here are **30 MCQs on the topic "Decoders"** for your **CUET PG MCA – Digital Fundamentals** preparation, including **CUET PG previous year questions (PYQs)** where applicable.

☒ **Topic: Decoders**

- ☒ Total Questions: 30
 - ☒ Correct answers are marked
 - ☒ PYQ = CUET PG Previous Year Question
-

1. What is the primary function of a decoder?

- a) Addition
 - b) Memory storage
 - ☒ c) Convert binary input to unique output
 - d) Multiplication
-

2. A decoder with n input lines can have a maximum of:

- a) $2n$ outputs
 - ☒ b) 2^n outputs
 - c) n^2 outputs
 - d) $n/2$ outputs
-

3. A 3-to-8 decoder has:

- ☒ a) 3 inputs and 8 outputs
 - b) 8 inputs and 3 outputs
 - c) 3 outputs and 8 inputs
 - d) 3 inputs and 4 outputs
-

4. Which logic function can a decoder implement?

- a) Parity
 - ☒ b) Minterms
 - c) Exponentiation
 - d) NAND
-

5. A 2-to-4 line decoder has how many output lines?

- a) 2
 - b) 8
 - ☒ c) 4
 - d) 1
-

6. (PYQ) The function of a 4-to-16 decoder is to:

- a) Convert BCD to binary
 - b) Add binary numbers
 - ☒ c) Generate 16 outputs for 4-bit input
 - d) Encode 16 inputs to 4 outputs
-

7. Which one is a standard IC for a 3-to-8 decoder?

- a) 7400
 - b) 7483
 - ☒ c) 74138
 - d) 74150
-

8. Which IC is used for BCD to Decimal decoding?

- a) 74138
 - ☒ b) 7442
 - c) 7485
 - d) 7402
-

9. How many outputs does a 4-to-16 decoder generate?

- a) 4
 - b) 8
 - ☒ c) 16
 - d) 10
-

10. What is an enable input in a decoder?

- ☒ a) Activates the decoder
 - b) Resets all outputs
 - c) Controls voltage
 - d) Multiplies inputs
-

11. How many 2-to-4 decoders are needed to make a 3-to-8 decoder?

- a) 2
 - b) 3
 - ☒ c) 2 (with enable inputs)
 - d) 1
-

12. The output of a decoder is:

- a) Always high
 - ☒ b) Only one line active at a time
 - c) All lines active
 - d) Based on OR gates
-

13. In digital systems, decoders are used for:

- a) Addition
 - b) Input buffering
 - ☐ c) **Address selection/memory decoding**
 - d) Multiplexing
-

14. Which logic gates are primarily used in decoders?

- ☐ a) **AND gates**
 - b) OR gates
 - c) XOR gates
 - d) NAND gates
-

15. A decoder circuit activates:

- a) All outputs
 - ☐ b) **Only one output for each input combination**
 - c) No output
 - d) Two outputs always
-

16. What is the Boolean expression for output Y_0 in a 2-to-4 decoder with inputs A and B?

- ☐ a) **$Y_0 = \bar{A} \cdot \bar{B}$**
 - b) $Y_0 = A \cdot B$
 - c) $Y_0 = A + B$
 - d) $Y_0 = \bar{A} + \bar{B}$
-

17. Decoders are essential in:

- ☐ a) **Memory address decoding**
 - b) Serial transmission
 - c) Voltage regulation
 - d) Frequency modulation
-

18. What does a BCD to Decimal decoder do?

- ☒ a) Converts binary coded decimal to decimal output
 - b) Converts decimal to binary
 - c) Multiplies BCD
 - d) Divides binary inputs
-

19. How many 3-to-8 decoders are needed to make a 4-to-16 decoder?

- a) 4
 - ☒ b) 2
 - c) 3
 - d) 5
-

20. A decoder with enable input is also called a:

- ☒ a) Controlled decoder
 - b) Binary counter
 - c) Comparator
 - d) Flip-flop
-

21. In a 3-to-8 decoder, when input is 101, which output line is high?

- a) Y3
 - b) Y1
 - ☒ c) Y5
 - d) Y7
-

22. Which of these is a BCD to 7-segment decoder IC?

- a) 7442
 - ☒ b) 7447
 - c) 7400
 - d) 7410
-

23. What is the function of IC 7447?

- a) BCD to decimal decoder
 - ☒ b) BCD to 7-segment display decoder
 - c) Parity generator
 - d) AND gate
-

24. In BCD, how many valid input combinations exist for a decoder?

- a) 16
 - ☒ b) 10
 - c) 8
 - d) 4
-

25. (PYQ) The decoder is used in:

- ☒ a) Microprocessor memory selection
 - b) Flip-flop triggering
 - c) Serial conversion
 - d) Shift register
-

26. Which of the following is NOT a use of a decoder?

- a) Data routing
 - b) Memory addressing
 - ☒ c) Data compression
 - d) 7-segment display
-

27. How many outputs will a 5-to-32 decoder have?

- a) 16
 - ☒ b) 32
 - c) 10
 - d) 64
-

28. A decoder circuit is designed using:

- a) Latches
 - b) Flip-flops
 - ☒ c) Logic gates
 - d) Counters
-

29. Which combination is best for building hierarchical decoders?

- ☒ a) Using enable pins
 - b) Using NANDs
 - c) Using counters
 - d) Using XOR
-

30. In a 4-to-16 decoder, how many select lines are there?

- a) 2
 - b) 4
 - ☒ c) 4
 - d) 16
-

Here are **30 MCQs on the topic "Multiplexers"** for your **CUET PG MCA - Digital Fundamentals** preparation, including **CUET PG previous year questions (PYQs)** wherever applicable.

☒ **Topic: Multiplexers**

☒ Total: 30 Questions

☒ Correct answers are marked

☒ PYQ = CUET PG Previous Year Question

1. What is the function of a multiplexer?

- a) Perform arithmetic operations
 - b) Convert binary to decimal
 - ☒ c) Select one input from many and forward it
 - d) Encode data
-

2. A 4-to-1 multiplexer has:

- ☒ a) 4 inputs, 2 selection lines, 1 output
 - b) 4 inputs, 1 selection line, 2 outputs
 - c) 2 inputs, 4 selection lines
 - d) 4 outputs, 1 input
-

3. Number of selection lines required for 8-to-1 multiplexer:

- a) 4
 - b) 2
 - ☒ c) 3
 - d) 1
-

4. A 2^n -to-1 multiplexer requires how many select lines?

- ☒ a) n
 - b) $2n$
 - c) n^2
 - d) $\log_2 n$
-

5. What is the Boolean expression for a 2-to-1 multiplexer?

- ☒ a) $Y = A \cdot \bar{S} + B \cdot S$
 - b) $Y = A + B$
 - c) $Y = A \cdot B$
 - d) $Y = A \oplus B$
-

6. (PYQ) Multiplexers are used for:

- ☒ a) Data routing
 - b) Code conversion
 - c) Memory decoding
 - d) Counting
-

7. What does MUX stand for?

- a) Maximum
 - ☒ b) **Multiplexer**
 - c) Multiplier
 - d) Multisystem
-

8. A 16-to-1 multiplexer has how many selection lines?

- a) 2
 - b) 3
 - ☒ c) **4**
 - d) 5
-

9. A multiplexer is also known as:

- a) Decoder
 - ☒ b) **Data selector**
 - c) Encoder
 - d) Demultiplexer
-

10. Which logic gate combination is used to build MUX?

- a) XOR and NOT
 - b) OR and NOT
 - ☒ c) **AND, OR, NOT**
 - d) NAND only
-

11. How many inputs does an 8-to-1 multiplexer have?

- a) 4
 - ☒ b) **8**
 - c) 3
 - d) 1
-

12. Which of the following IC is used for 4-to-1 MUX?

- a) 74138
 - ☒ b) 74153
 - c) 7483
 - d) 7400
-

13. MUX can be used to implement:

- ☒ a) Any Boolean function
 - b) Only SOP expressions
 - c) Flip-flops
 - d) Counters
-

14. Which is true for MUX?

- ☒ a) 1 output line
 - b) 1 input and multiple outputs
 - c) Only 1 select line
 - d) No logic gates are used
-

15. Which of the following is an application of MUX?

- ☒ a) Digital signal routing
 - b) Voltage amplification
 - c) Frequency modulation
 - d) Data encryption
-

16. Which of the following is true about a multiplexer?

- a) It demultiplexes signals
 - ☒ b) It selects one input out of many
 - c) It decodes signals
 - d) It acts as a timer
-

17. A MUX with 'm' selection lines can select how many inputs?

- a) m
 - ☒ b) 2^m
 - c) $2m$
 - d) m^2
-

18. What is the minimum number of 2-to-1 MUX needed to build an 8-to-1 MUX?

- a) 4
 - b) 6
 - ☒ c) 7
 - d) 8
-

19. Which IC is used for 8-to-1 multiplexer?

- a) 74138
 - ☒ b) 74151
 - c) 7485
 - d) 74157
-

20. (PYQ) The output of a multiplexer is determined by:

- a) Input values only
 - ☒ b) Select lines
 - c) Number of gates
 - d) Carry lookahead
-

21. Which of the following is NOT a feature of multiplexer?

- a) Can select one input
 - b) Has multiple selection lines
 - ☒ c) Has multiple outputs
 - d) Used in communication
-

22. The control inputs of a multiplexer are also called:

- a) Data inputs
 - ☒ b) Select lines
 - c) Address bus
 - d) Clock lines
-

23. A 1-to-2 line is called:

- a) Multiplexer
 - b) Encoder
 - c) Selector
 - ☒ d) Demultiplexer
-

24. Can a multiplexer perform logic function implementation?

- ☒ a) Yes, any Boolean function
 - b) Only XOR
 - c) Only NOT
 - d) No
-

25. MUX is a combination of:

- a) Sequential circuits
 - ☒ b) Combinational circuits
 - c) Memory devices
 - d) Flip-flops
-

26. To design a 32-to-1 MUX, how many select lines are required?

- a) 4
 - b) 5
 - ☒ c) 5
 - d) 6
-

27. A 2-to-1 MUX uses how many data inputs?

- ☐ a) 2
 - b) 1
 - c) 3
 - d) 4
-

28. (PYQ) MUX circuits are useful in:

- a) Arithmetic operations
 - ☐ b) **Channel selection in communication**
 - c) Decoding BCD
 - d) Flip-flop triggering
-

29. How many total inputs and outputs are there in 4-to-1 MUX (excluding enable)?

- ☐ a) **4 inputs, 1 output**
 - b) 2 inputs, 4 outputs
 - c) 1 input, 4 outputs
 - d) 4 inputs, 4 outputs
-

30. Which of these is NOT a standard MUX IC?

- a) 74151
 - b) 74153
 - ☐ c) **74123**
 - d) 74157
-

Here are **30 MCQs on "Registers and Counters"** for your **CUET PG MCA – Digital Fundamentals** preparation, including **previous CUET PG questions (PYQs)** where available.

☐ **Topic: Registers and Counters**

☐ Total: 30 Questions

☐ Correct answers marked

☐ PYQ = CUET PG Previous Year Question

1. What is a register in digital electronics?

- ☒ a) Group of flip-flops for data storage
 - b) Clock generator
 - c) Logic gate
 - d) Counter circuit
-

2. A 4-bit register can store:

- a) 8 values
 - b) 2 values
 - ☒ c) 4 bits of data
 - d) 16 bytes
-

3. A counter is a:

- ☒ a) Sequential circuit
 - b) Combinational circuit
 - c) Decoder
 - d) Encoder
-

4. (PYQ) The main function of a counter is:

- a) Logic simplification
 - ☒ b) Count clock pulses
 - c) Convert analog to digital
 - d) Store bytes
-

5. Which of the following is used for data transfer?

- a) Decoder
 - ☒ b) Register
 - c) Counter
 - d) ALU
-

6. Which is not a type of counter?

- a) Synchronous
 - b) Asynchronous
 - ☐ c) **Analog**
 - d) Ring
-

7. A register which shifts bits left or right is called:

- a) Ring register
 - ☐ b) **Shift register**
 - c) Parallel register
 - d) Toggle register
-

8. In asynchronous counter, flip-flops are:

- a) Clocked simultaneously
 - ☐ b) **Clocked by previous flip-flop**
 - c) Not used
 - d) Replaced with latches
-

9. A 3-bit counter can count from:

- a) 0 to 3
 - ☐ b) **0 to 7**
 - c) 1 to 8
 - d) 0 to 15
-

10. Which register stores data temporarily during CPU operations?

- ☐ a) **Accumulator**
 - b) Decoder
 - c) MAR
 - d) ALU
-

11. (PYQ) What is the MOD number of a 4-bit binary counter?

- a) 8
 - ☒ b) 16
 - c) 10
 - d) 4
-

12. Which counter counts both up and down?

- a) Ripple counter
 - ☒ b) Up-down counter
 - c) Ring counter
 - d) T flip-flop
-

13. Which is faster?

- a) Asynchronous counter
 - ☒ b) Synchronous counter
 - c) Both same
 - d) None
-

14. How many flip-flops are needed for MOD-16 counter?

- a) 3
 - ☒ b) 4
 - c) 2
 - d) 5
-

15. A shift register used to convert serial to parallel is:

- ☒ a) SIPO
 - b) PISO
 - c) SISO
 - d) PIPO
-

16. In a ring counter, the output of the last FF is connected to:

- ☒ a) First flip-flop input
 - b) Ground
 - c) Clock
 - d) Output bus
-

17. A 4-bit bidirectional shift register can:

- a) Shift only left
 - b) Shift only right
 - ☒ c) Shift both directions
 - d) Store 8 bits
-

18. A Johnson counter is also called:

- ☒ a) Twisted ring counter
 - b) MOD counter
 - c) Up counter
 - d) Ripple counter
-

19. Which register is used in UART for serial data?

- a) Program counter
 - ☒ b) Shift register
 - c) Ring counter
 - d) Decoder
-

20. A register that loads all bits at once is:

- a) Serial register
 - ☒ b) Parallel register
 - c) Ring register
 - d) Decoder
-

21. Which counter resets automatically after its MOD count?

- ☐ a) MOD-n counter
 - b) Synchronous
 - c) Down counter
 - d) Up-down counter
-

22. Flip-flop used in most counters:

- a) D flip-flop
 - b) JK flip-flop
 - ☐ c) T flip-flop
 - d) SR flip-flop
-

23. What is the function of a counter in timers?

- a) Store data
 - ☐ b) Generate delay
 - c) Add numbers
 - d) Encode data
-

24. In a PIPO register, data is:

- a) Serial in, parallel out
 - b) Parallel in, serial out
 - ☐ c) Parallel in, parallel out
 - d) None
-

25. The output of shift register is delayed by:

- a) 1 cycle
 - b) n cycles
 - ☐ c) Clock pulses
 - d) Flip-flop delay
-

26. How many states in MOD-6 counter?

- ☒ a) 6
 - b) 64
 - c) 16
 - d) 10
-

27. A 3-stage ring counter has how many states?

- a) 2
 - ☒ b) 3
 - c) 6
 - d) 8
-

28. A register that moves bits in a circular path is:

- a) Shift register
 - b) Ripple counter
 - ☒ c) Ring counter
 - d) PISO
-

29. Main use of counters in digital systems is:

- a) Decoding
 - b) Arithmetic
 - ☒ c) Timing and control
 - d) Multiplexing
-

30. (PYQ) A 4-bit synchronous counter counts from:

- a) 0 to 3
 - b) 1 to 10
 - ☒ c) 0 to 15
 - d) 0 to 10
-

Here are 30 MCQs on "Memory Unit" for your CUET PG MCA - Digital Fundamentals preparation, including previous CUET PG questions (marked as PYQ).

☐ **Topic: Memory Unit**

☐ Total Questions: 30

☐ Correct answers marked

☐ PYQs marked where applicable

1. Memory unit is a part of:

a) ALU

b) CU

☐ **c) CPU**

d) I/O Unit

2. Which of the following is a volatile memory?

a) ROM

☐ **b) RAM**

c) Flash

d) EEPROM

3. What type of memory is used to store the BIOS?

☐ **a) ROM**

b) RAM

c) DRAM

d) Cache

4. Which memory stores data permanently?

☐ **a) ROM**

b) RAM

c) Cache

d) DRAM

5. The fastest memory in the computer is:

- a) RAM
 - ☒ b) Cache
 - c) Hard disk
 - d) ROM
-

6. Which is non-volatile memory?

- ☒ a) ROM
 - b) DRAM
 - c) SRAM
 - d) Cache
-

7. (PYQ) A memory that can be both read and written is:

- a) ROM
 - ☒ b) RAM
 - c) PROM
 - d) EPROM
-

8. Which of the following is used as main memory?

- a) ROM
 - ☒ b) RAM
 - c) Registers
 - d) Cache
-

9. Dynamic RAM (DRAM) needs to be:

- a) Rewritten
 - ☒ b) Refreshed
 - c) Burned
 - d) Deleted
-

10. What does EEPROM stand for?

- ☒ a) Electrically Erasable Programmable Read Only Memory
 - b) Electronically Enabled Programmable RAM
 - c) Erasable Electronic Programmable RAM
 - d) None
-

11. Static RAM (SRAM) is:

- ☒ a) Faster than DRAM
 - b) Slower than DRAM
 - c) Equal to DRAM
 - d) Non-volatile
-

12. Which memory is used for temporary storage during program execution?

- a) ROM
 - ☒ b) RAM
 - c) HDD
 - d) SSD
-

13. Cache memory is located between:

- ☒ a) CPU and RAM
 - b) RAM and Hard disk
 - c) ROM and RAM
 - d) ALU and CU
-

14. The smallest unit of memory is:

- ☒ a) Bit
 - b) Byte
 - c) Word
 - d) Nibble
-

15. A memory unit that stores data in fixed locations is:

- ☒ a) ROM
 - b) DRAM
 - c) SRAM
 - d) Cache
-

16. PROM stands for:

- ☒ a) Programmable Read-Only Memory
 - b) Permanent ROM
 - c) Protected ROM
 - d) Programmed RAM
-

17. The memory directly accessed by the CPU is:

- ☒ a) Main memory (RAM)
 - b) Hard disk
 - c) SSD
 - d) Optical drive
-

18. (PYQ) Which memory is used to increase system speed temporarily?

- a) ROM
 - ☒ b) Cache
 - c) CD-ROM
 - d) USB
-

19. Which memory can be programmed once and not erased?

- ☒ a) PROM
 - b) EPROM
 - c) EEPROM
 - d) DRAM
-

20. Memory used for virtual memory system:

- a) RAM only
 - ☒ b) RAM + Hard disk
 - c) ROM only
 - d) EEPROM
-

21. Access time in memory means:

- a) Delay in clock
 - ☒ b) Time to read/write data
 - c) CPU processing time
 - d) Transfer rate
-

22. What is the size of a nibble?

- ☒ a) 4 bits
 - b) 8 bits
 - c) 16 bits
 - d) 2 bits
-

23. Which of the following is not a secondary memory?

- ☒ a) RAM
 - b) CD
 - c) HDD
 - d) SSD
-

24. A 64 KB memory means:

- a) 64×10^3 bits
 - ☒ b) 64×1024 bytes
 - c) 64×1000 bytes
 - d) 64×1024 bits
-

25. Memory used to store bootloader is:

- ☒ a) ROM
 - b) RAM
 - c) HDD
 - d) DRAM
-

26. The term “memory hierarchy” is based on:

- a) Speed only
 - ☒ b) Speed, cost, size
 - c) Cost only
 - d) Power consumption
-

27. Flash memory is a type of:

- ☒ a) Non-volatile memory
 - b) Volatile memory
 - c) DRAM
 - d) Cache
-

28. Which memory has the longest access time?

- a) Cache
 - b) RAM
 - ☒ c) Hard Disk
 - d) ROM
-

29. Memory addressing is handled by:

- ☒ a) Memory Address Register (MAR)
 - b) Memory Data Register (MDR)
 - c) Control Unit
 - d) ALU
-

30. Which of these is used to transfer data from memory to processor?

a) MAR

☐ b) **MDR**

c) IR

d) PC
